Large-Scale Dataset Analysis: 250K Job Records

March 29, 2026

1 Large-Scale Dataset Analysis: 250K Job Records

1.1 Project Overview

```
<h2 style="color: #4db8ff;">Large-Scale Dataset Analysis: 250K Job Records</h2>
<p>We analyzed 250,000 jobs in the fields of: <b>'AI Engineer', 'Data Analyst', 'Frontend Devel
<p>In this project, we analyzed data downloaded from <b>Kaggle</b>. We started the project by l
<h3 style="color: #ffcc00;">Part 1: Data cleaning, memory optimization and validation</h3>
<p>In this part, we looked for potential issues, such as duplicate strings in different formats
<p>We added an important part under the title: <b>"Beyond the Numbers: How Realistic is this Sa
<p>Finally, in this part, we added a title <b>"Data Validation"</b> to define the analysis metl
<h3 style="color: #ffcc00;">Part 2: Analysis</h3>
We performed four main parts of the analysis.
<ul>
    <li>Correlation Analysis: Years of Experience vs. Annual Salary</li>
    <li>Analyzing the Correlation of Education Level with Salary</li>
    <li>Country-wise Comparison</li>
    <li>Job title comparison</li>
</ul>
<h3 style="color: #ffcc00;">Part 3: Is the data synthetic?</h3>
<p>In the third and final part of the project under the title <b>"Is the data synthetic?"</b>,
```

1.2 Dependencies & Data Import

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
import seaborn as sns
import numpy as np
```

[2]:
```
df = pd.read_csv('../Dataset/original_dataset.csv')
```

This dataset is under Creative Commons CC0 (Public Domain Dedication).

1.3 I. Data cleaning, memory optimization and validation

[3]:
```
df.head(10)
```

[3]:

```
                   job_title  experience_years education_level  skills_count  \
0                AI Engineer                10        Bachelor             2
1               Data Analyst                 5        Bachelor            17
2         Frontend Developer                18             PhD             4
3           Business Analyst                19             PhD            13
4            Product Manager                15        Bachelor             7
5                AI Engineer                 0     High School             4
6          Backend Developer                 6     High School            16
7  Machine Learning Engineer                 4        Bachelor            18
8                AI Engineer                 5         Diploma            14
9         Frontend Developer                18             PhD             2

        industry company_size   location remote_work  certifications  salary
0     Healthcare       Medium      India      Hybrid               2  109413
1        Telecom        Small  Australia          No               0   93764
2          Media       Medium  Singapore          No               1  148123
3         Retail       Medium     Canada         Yes               0  189123
4  Manufacturing        Large     Sweden         Yes               0  165069
5      Education   Enterprise        USA          No               2  180351
6  Manufacturing   Enterprise     Canada          No               3  165375
7     Healthcare   Enterprise     Canada      Hybrid               5  202463
8        Telecom   Enterprise  Singapore         Yes               0  171834
9         Retail   Enterprise      India      Hybrid               5  128377
```

In the previous cell, we used `df.head(10)` to display the first 10 rows of the DataFrame. This allows us to quickly inspect the structure of the data.

1.3.1 I-1. Data Structure & Technical Audit

[4]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   job_title         250000 non-null  str
 1   experience_years  250000 non-null  int64
 2   education_level   250000 non-null  str
 3   skills_count      250000 non-null  int64
 4   industry          250000 non-null  str
 5   company_size      250000 non-null  str
 6   location          250000 non-null  str
 7   remote_work       250000 non-null  str
 8   certifications    250000 non-null  int64
 9   salary            250000 non-null  int64
dtypes: int64(4), str(6)
memory usage: 30.7 MB
```

A technical inspection of the 250,000 records using df.info() reveals the following:

- Data Integrity
  The dataset is fully complete with no missing values, meaning no imputation is required at this stage.
- Feature Types
  The dataset includes 4 numerical columns (e.g., salary, experience_years) correctly stored as int64, and 6 categorical columns currently represented as strings.
- Memory Optimization
  The dataset currently uses approximately 30.7 MB of memory.
  To improve efficiency, high-frequency string columns such as industry, education_level, and remote_work will be converted to the category data type, reducing memory usage.

1.3.2 I-2. Categorical Values Validation (Data Integrity Check)

[5]:
```
text_cols = df.select_dtypes(include=['object']).columns

for col in text_cols:
    print(f"--- Unique values in {col} ---")
    print(df[col].unique())
    print("\n")
```

```
--- Unique values in job_title ---
<ArrowStringArray>
[              'AI Engineer',              'Data Analyst',
        'Frontend Developer',          'Business Analyst',
           'Product Manager',         'Backend Developer',
 'Machine Learning Engineer',           'DevOps Engineer',
         'Software Engineer',     'Cybersecurity Analyst',
            'Data Scientist',            'Cloud Engineer']
Length: 12, dtype: str


--- Unique values in education_level ---
<ArrowStringArray>
['Bachelor', 'PhD', 'High School', 'Diploma', 'Master']
Length: 5, dtype: str


--- Unique values in industry ---
<ArrowStringArray>
[   'Healthcare',       'Telecom',         'Media',        'Retail',
 'Manufacturing',     'Education',       'Finance',    'Technology',
    'Consulting',    'Government']
Length: 10, dtype: str
```

```
--- Unique values in company_size ---
<ArrowStringArray>
['Medium', 'Small', 'Large', 'Enterprise', 'Startup']
Length: 5, dtype: str


--- Unique values in location ---
<ArrowStringArray>
[      'India',   'Australia',   'Singapore',      'Canada',      'Sweden',
         'USA', 'Netherlands',      'Remote',     'Germany',          'UK']
Length: 10, dtype: str


--- Unique values in remote_work ---
<ArrowStringArray>
['Hybrid', 'No', 'Yes']
Length: 3, dtype: str



/tmp/ipykernel_14315/2293762028.py:1: Pandas4Warning: For backward
compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is
specified. This behavior is deprecated and will be removed in a future version.
Explicitly pass 'str' to `include` to select them, or to `exclude` to remove
them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-
migration-select-dtypes for details on how to write code that works with pandas
2 and 3.
  text_cols = df.select_dtypes(include=['object']).columns
```

After performing a deep dive into the unique values of each categorical column, we have verified the following: 1. Data Consistency & Cleaning Quality

```
No Redundancy: There are no duplicate categories written in different styles (e.g., "Remote" an

No Whitespace Issues: All string values are "trimmed" and clean; there are no leading or traili


Status:  Production-Ready

Conclusion: Everything is perfectly aligned. The categories are logical, standardized, and clea
```

1.3.3 I-3. Statistical Summary & Sanity Check

[6]:
```
df.describe()
```

[6]:

	experience_years	skills_count	certifications	salary
count	250000.000000	250000.000000	250000.000000	250000.000000
mean	10.005408	9.997812	2.491928	145718.080524

```
std            6.060602       5.479288        1.706475   37407.952729
min            0.000000       1.000000        0.000000   31867.000000
25%            5.000000       5.000000        1.000000  119358.000000
50%           10.000000      10.000000        2.000000  143453.000000
75%           15.000000      15.000000        4.000000  169492.000000
max           20.000000      19.000000        5.000000  333046.000000
```

Through the analysis of Descriptive Statistics for numerical features:

- Range: Salaries range from $31,867 (Minimum) to $333,046 (Maximum).
- There are no "impossible" values (such as negative salaries, 1000 years of experience, or millions of certificates).

  Status : There is still an underlying issue that we will address in the section: I-7. Beyond the Numbers: How Realistic is this Salary Data

1.3.4 I-4. Initial Memory Consumption (Pre-Optimization)

[7]:
```python
print(df.memory_usage(deep=True))
```

```
Index                  132
job_title          6061551
experience_years   2000000
education_level    3750430
skills_count       2000000
industry           4174893
company_size       3648620
location           3600551
remote_work        2916807
certifications     2000000
salary             2000000
dtype: int64
```

Before applying any data type transitions, we inspected the memory footprint of the DataFrame using df.memory_usage(deep=True). 1. Analysis of Current State

```
Heavyweight Columns: The text-based columns (stored as object) are consuming the most memory. I

    job_title: ~6.06 MB

    industry: ~4.17 MB

    education_level: ~3.75 MB
```

2. Technical Observation

   Status: Pre-Optimization Phase

   Issue: Storing repetitive strings as "Objects" is inefficient. Each entry is being stored as a full string pointer, which scales poorly as the dataset grows.

Observation: The total memory usage is dominated by these categorical strings, which are prime candidates for optimization.

3. Transition Goal

This baseline measurement serves as our Control Group. In the next step, we will convert these 6 string columns to the category type to observe the drastic reduction in memory overhead.

1.3.5 I-5. Data Type Transformation (Memory Optimization)

[8]:
```
df['job_title'] = df['job_title'].astype('category')
df['education_level'] = df['education_level'].astype('category')
df['industry'] = df['industry'].astype('category')
df['company_size'] = df['company_size'].astype('category')
df['location'] = df['location'].astype('category')
df['remote_work'] = df['remote_work'].astype('category')
```

In this step, we executed a strategic conversion of categorical features to enhance performance and reduce memory overhead. 1. The Engineering Action

We identified 6 columns originally stored as generic Strings (Objects) and explicitly cast them to the category data type:

`job_title, education_level, industry, company_size, location, and remote_work.`

2. Why Category over String?

   Storage Efficiency: Instead of storing the full text for each of the 250,000 rows, Pandas now stores a unique integer key for each category, mapping it to the original string in a small lookup table.

   Computational Speed: Operations such as sorting, grouping, and unique value counting are now significantly faster as they operate on integers "under the hood."

[9]:
```
print(df.memory_usage(deep=True))
```

```
Index                   132
job_title            250293
experience_years    2000000
education_level      250076
skills_count        2000000
industry             250169
company_size         250074
location             250146
remote_work          250036
certifications      2000000
salary              2000000
dtype: int64
```

Status: Optimized

`Result: The data remains logically identical, but the internal representation is now optimized`

[10]:
```
## df.to_parquet('jobs_data_optimized.parquet')
```

After optimizing the data types and verifying the integrity of the 250,000 records, we exported the final DataFrame to a Parquet file format.

[11]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   job_title         250000 non-null  category
 1   experience_years  250000 non-null  int64
 2   education_level   250000 non-null  category
 3   skills_count      250000 non-null  int64
 4   industry          250000 non-null  category
 5   company_size      250000 non-null  category
 6   location          250000 non-null  category
 7   remote_work       250000 non-null  category
 8   certifications    250000 non-null  int64
 9   salary            250000 non-null  int64
dtypes: category(6), int64(4)
memory usage: 9.1 MB
```

1.3.6 I-6. Optimization Results

After converting the string-based columns to the category data type, we re-inspected the DataFrame structure to quantify the performance gains. 1. Key Observations

```
Data Type Update: All 6 target columns (job_title, education_level, etc.) are now successfully
```

2. Memory Usage Impact

   Initial Memory: ~30.7 MB

   Optimized Memory: 9.1 MB

   Efficiency Gain: We achieved a ~70% reduction in memory footprint.

3. Why this matters for Scalability

By reducing the memory usage to less than a third of its original size, we have:

```
Faster Computations: Operations like df.groupby() or df.pivot_table() will now run much faster
```

1.3.7 I-7. Beyond the Numbers: How Realistic is this Salary Data

[12]:
```
office_based_df = df[df['remote_work'] == 'No']

job_location_median = office_based_df.pivot_table(
    index='job_title',
```

```
    columns='location',
    values='salary',
    aggfunc='median'
).fillna(0)

print("Median Salaries by Job Title and Location (Office-Based Only):")
display(job_location_median)
```

```
Median Salaries by Job Title and Location (Office-Based Only):

location                    Australia    Canada   Germany     India  \
job_title
AI Engineer                  160215.0  192125.0  181225.0  112848.0
Backend Developer            130423.5  157828.0  141210.0   90420.5
Business Analyst             114531.5  134063.5  127232.0   79181.0
Cloud Engineer               141148.0  169855.0  155653.0   98713.0
Cybersecurity Analyst        140484.0  166226.0  151249.0   95770.0
Data Analyst                 110535.0  133458.0  122220.0   78205.5
Data Scientist               136338.5  164743.0  150207.0   95579.0
DevOps Engineer              140830.0  167385.0  155249.0   98638.0
Frontend Developer           123114.0  148758.0  135843.5   87090.0
Machine Learning Engineer    154571.0  181855.0  168476.5  106114.0
Product Manager              146159.0  176583.5  161170.0  102980.0
Software Engineer            128681.0  156109.5  146192.0   93570.0

location                    Netherlands    Remote  Singapore    Sweden  \
job_title
AI Engineer                    164525.0  162693.0   158868.0  162783.0
Backend Developer              129058.0  129841.0   128358.0  129308.0
Business Analyst               114380.0  112881.0   113750.0  114907.5
Cloud Engineer                 142053.0  142386.0   144156.5  142248.0
Cybersecurity Analyst          137450.5  138998.5   138872.0  139495.0
Data Analyst                   112271.0  111394.0   110897.0  111186.0
Data Scientist                 136546.0  137947.0   137920.0  135681.0
DevOps Engineer                140120.5  141713.0   139578.5  140751.0
Frontend Developer             122473.0  124232.0   125362.5  123380.0
Machine Learning Engineer      150847.5  152489.5   151614.0  153472.0
Product Manager                147222.0  147296.0   147203.0  147679.0
Software Engineer              130121.0  130353.0   130999.0  132997.0

location                          UK       USA
job_title
AI Engineer                 185974.0  211324.0
Backend Developer           149664.0  170533.0
Business Analyst            133038.0  146194.0
Cloud Engineer              161609.0  185816.0
Cybersecurity Analyst       159397.0  183884.5
Data Analyst                130915.0  147261.0
```

```
Data Scientist              159105.0  178626.0
DevOps Engineer             162482.0  181742.0
Frontend Developer          142595.0  164401.0
Machine Learning Engineer   176169.0  198663.5
Product Manager             167674.0  191914.5
Software Engineer           152779.0  171179.0
```

This table illustrates the Median Salary per country and job title for office-based roles **(Remote Work = No)**.

Let's take India as a sample from this table. The median salary shown for India does not seem to reflect reality; however, this could be due to a small sample size, which can often be misleading. Let's verify:

[13]:
```python
office_based_df = df[df['remote_work'] == 'No']

job_location_count = office_based_df.pivot_table(
    index='job_title',
    columns='location',
    values='salary',
    aggfunc='count'
).fillna(0)

print("Sample Size (Count) per Job and Location (Office-Based Only):")
display(job_location_count)
```

```
Sample Size (Count) per Job and Location (Office-Based Only):

location                   Australia  Canada  Germany  India  Netherlands  \
job_title
AI Engineer                      672     715      662    689          689
Backend Developer                704     677      745    712          698
Business Analyst                 720     692      731    691          663
Cloud Engineer                   670     707      705    667          675
Cybersecurity Analyst            735     697      729    753          684
Data Analyst                     677     705      697    744          693
Data Scientist                   728     711      685    659          692
DevOps Engineer                  717     711      688    691          712
Frontend Developer               711     691      684    707          709
Machine Learning Engineer        700     742      668    715          650
Product Manager                  772     718      696    697          661
Software Engineer                671     702      649    645          687

location                   Remote  Singapore  Sweden   UK  USA
job_title
AI Engineer                   672        710     668  693  732
Backend Developer             680        671     702  681  695
Business Analyst              632        667     712  716  659
Cloud Engineer                642        714     681  704  731
Cybersecurity Analyst         702        693     712  675  664
```

```
Data Analyst                  713        701     670  709  678
Data Scientist                711        688     702  635  695
DevOps Engineer               692        718     695  729  696
Frontend Developer            671        696     698  686  669
Machine Learning Engineer     708        700     677  729  716
Product Manager               703        736     731  734  708
Software Engineer             708        666     722  703  753
```

This table displays the sample size (count) of salary records per country and job title for office-based roles (Remote Work = No).

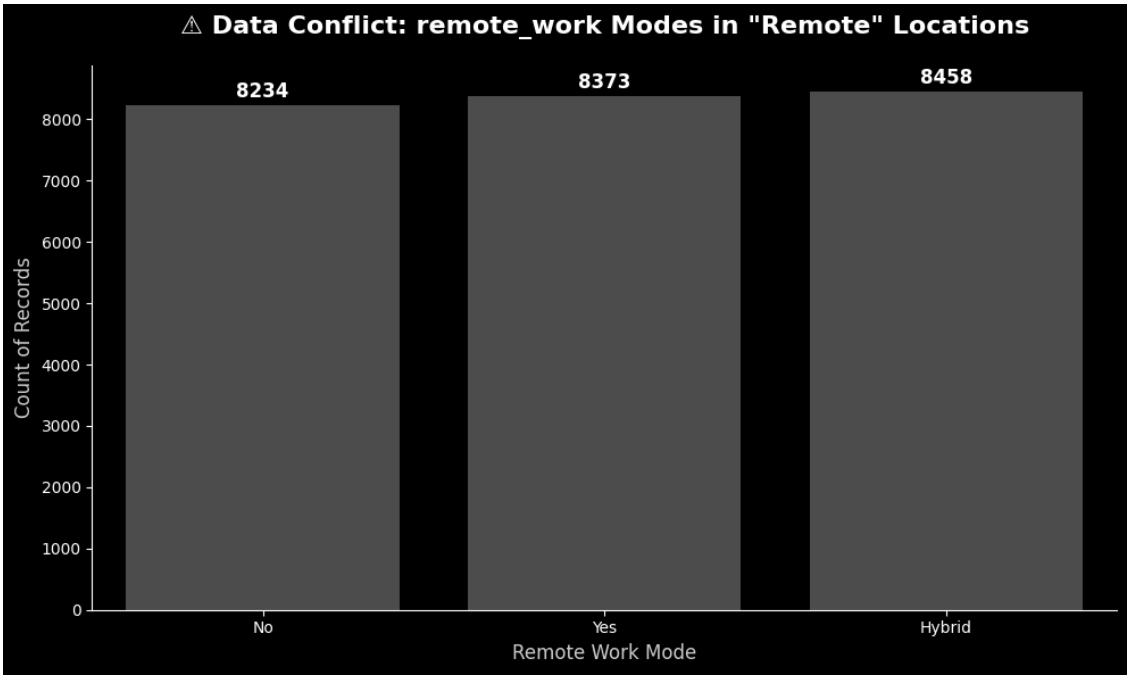
Upon observation, we have a sufficient sample size for job roles in India. Therefore, the unrealistic median salary is not because to a lack of data points. Instead, this suggests two possibilities:

```
- Selection Bias: The dataset may have been sampled from specific high-paying tiers

- Synthetic Nature: The data might be synthetic (artificial) and not reflective of real-world l
```

```
<h3 style="color: #cc0000; margin-top: 0;">  Critical Observation: Logical Data Conflict</h3>
<p style="color: #333;">
    Even after filtering the dataset for office-based roles <b>remote_work = No</b>, a signific
   This represents a logical inconsistency in the data. Which further confirms that this datase
</p>
```

[14]:
```
plt.style.use('dark_background')
plt.figure(figsize=(10, 6))

remote_location_df = df[df['location'] == 'Remote']
remote_work_counts = remote_location_df['remote_work'].value_counts().
 ↪reindex(['No', 'Yes', 'Hybrid'], fill_value=0)

ax = sns.barplot(x=remote_work_counts.index, y=remote_work_counts.
 ↪values,color="#fff", alpha=0.3)

plt.title('  Data Conflict: remote_work Modes in "Remote" Locations',
          fontsize=16, fontweight='bold', color='#ffffff', pad=20)
plt.xlabel('Remote Work Mode', fontsize=12, color='#cccccc')
plt.ylabel('Count of Records', fontsize=12, color='#cccccc')

sns.despine()

for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}',
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='baseline',
                fontsize=12, fontweight='bold', color='white', xytext=(0, 5),
                textcoords='offset points')

plt.tight_layout()
```

```
plt.show()
```



[15]:
```
%%html
<div style="padding: 20px;
            background-color: #1a1a1a;
            border-left: 6px solid #ff4d4d;
            border-radius: 8px;
            color: #ffffff;
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;">

    <h3 style="color: #ff4d4d; margin-top: 0; display: flex; align-items:␣
↪center;">
        <span style="margin-right: 10px;"> </span> Critical Alert
    </h3>

    <p style="font-size: 1.1em; line-height: 1.6; margin-bottom: 10px;">
        As this graph illustrates, there is a clear <b>inconsistency</b>: many␣
↪records labeled as
        <span style="color: #00ffcc; font-family: monospace;">"Remote"</span>␣
↪in the location column are
        simultaneously categorized as <span style="color: #ffcc00; font-family:␣
↪monospace;">"No"</span>
        for remote work.
    </p>
```

```
    <p style="font-size: 1.1em; line-height: 1.6; margin-bottom: 10px;">
        This confirms a <b>logical discrepancy</b> in the data, strongly␣
↪suggesting that the dataset is
        <span style="text-decoration: underline; color: #ff4d4d;">synthetic or␣
↪artificially generated</span>.
    </p>

    <hr style="border: 0; border-top: 1px solid #444; margin: 15px 0;">


</div>
```

<IPython.core.display.HTML object>

1.3.8 I-8. Data Validation

[16]:
```
%%html
<div style="padding: 15px;
            background-color: #f8f9fa;
            border-left: 5px solid #2ecc71;
            border-radius: 5px;
            color: #2c3e50;
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;">

    <h4 style="color: #27ae60; margin-top: 0;">  Methodological Note: Numerical␣
↪Scope & Potential Bias</h4>

    <p style="font-size: 1em; line-height: 1.5;">
        In this analysis, we treat the data as <b>pure numerical values</b>␣
↪within the scope of this specific dataset. It is crucial to acknowledge that␣
↪direct cross-country comparisons may be misleading due to <b>the presence of␣
↪variables</b>.
    </p>

    <p style="font-size: 0.95em; line-height: 1.5; color: #34495e;">
        <i>Example:</i> We might have a country with a high average salary␣
↪where most of the data pertains to high school diplomas, while a country␣
↪with a low average salary where most of the data pertains to doctorates.

        This might not apply to these data points, but it's just an example.
    </p>

    <hr style="border: 0; border-top: 1px solid #ddd; margin: 10px 0;">

</div>
```

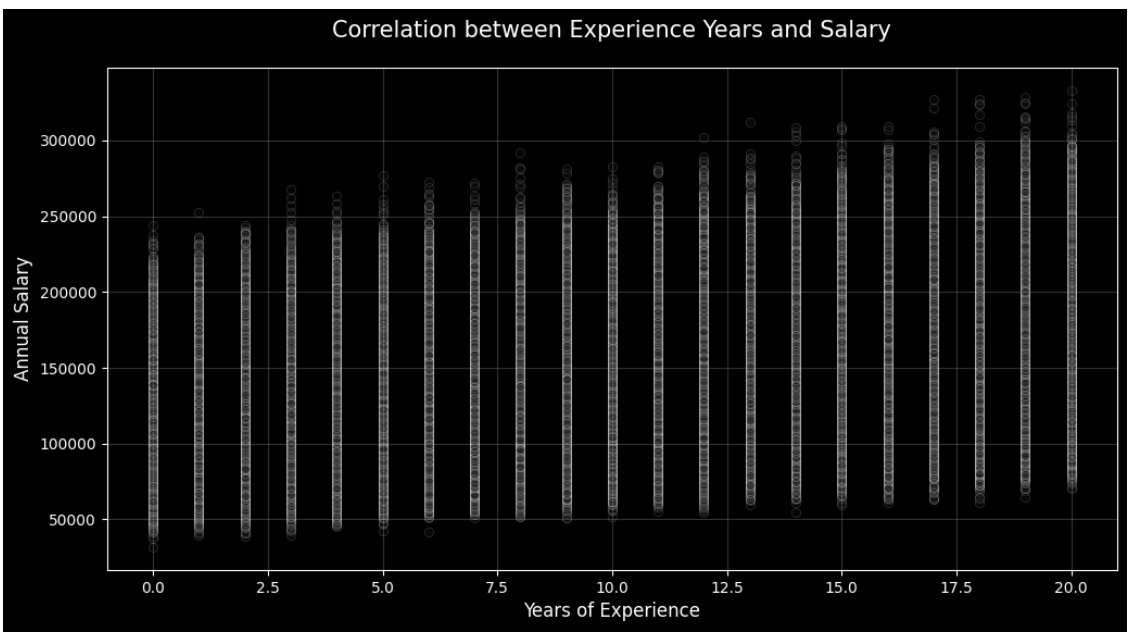
<IPython.core.display.HTML object>

1.4 II. Exploratory Data Analysis (EDA)

1.4.1 II-1. Correlation Analysis: Years of Experience vs. Annual Salary

I-1-1. Comprehensive Salary Mapping: A 250K-Record Scatter Analysis by years of experience

[17]:
```
plt.style.use('dark_background')
plt.figure(figsize=(12, 6))

sns.scatterplot(data=df, x='experience_years', y='salary', alpha=0.2,␣
 ↪color='#222')

plt.title('Correlation between Experience Years and Salary', fontsize=15,␣
 ↪pad=20)
plt.xlabel('Years of Experience', fontsize=12)
plt.ylabel('Annual Salary', fontsize=12)

plt.grid(True, alpha=0.2)
plt.show()
```



Based on the visualized 250,000 records in Correlation between Experience Years and Salary scatter plot, we can observe the following:

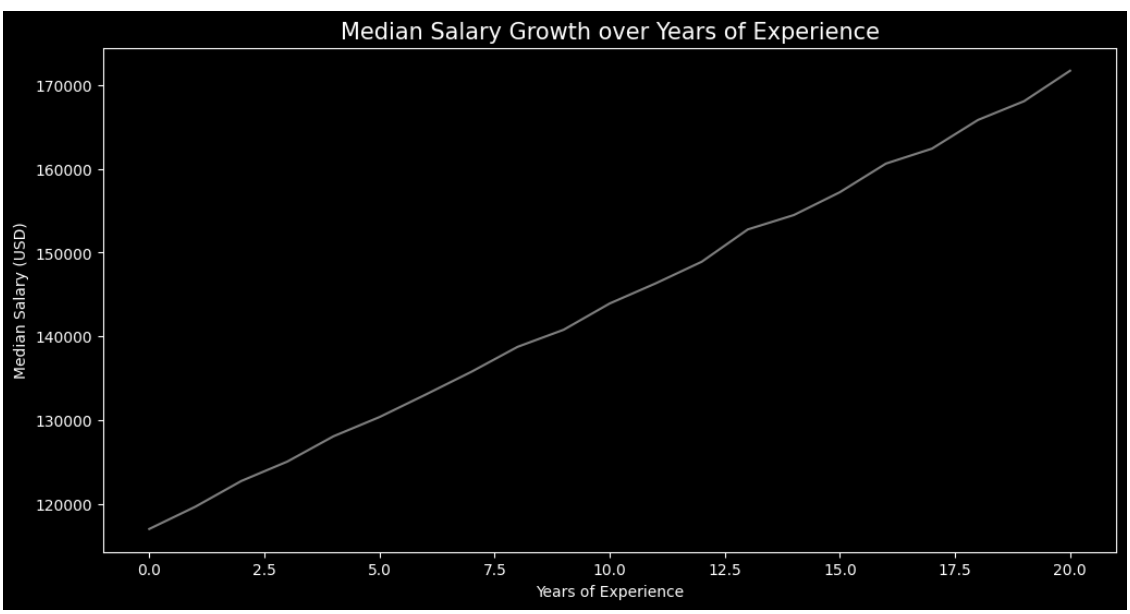
- Upward Trend in Max and Min Salaries: There is a clear positive correlation where an increase in Years of Experience leads to a consistent rise in both the Minimum and Maximum salary levels within the dataset.
- Shift in Density Concentration: While the overall range expands, we observe that the highest density of points (the darkest areas in the plot) gradually shifts upward as experience

increases. This indicates that the "market standard" for a typical salary moves to a higher baseline with seniority.

- Salary Overlap: The data shows that certain individuals with fewer years of experience earn higher salaries than others with significantly more seniority, suggesting that other factors (skills,industry or country) play a major role.
- Next Step: To gain a clearer understanding of the typical career progression, we will conduct a comparison of Median Salaries across all years of experience in the data.

II-1-2. Median Salary Growth over Years of Experience

[18]:
```
plt.style.use('dark_background')
plt.figure(figsize=(12, 6))

sns.lineplot(data=df, x='experience_years', y='salary', estimator=np.median,␣
 ↪errorbar=None, color='#fff', alpha=0.5)

plt.title('Median Salary Growth over Years of Experience', fontsize=15)
plt.xlabel('Years of Experience')
plt.ylabel('Median Salary (USD)')

plt.show()
```



Based on the Median Salary Growth over Years of Experience plot, we can observe the following:

```
Consistent Median Increase: There is a clear and steady upward trend in the median salary as ye
```

II-1-3. Year-over-Year Salary Growth: Absolute and Percentage Analysis

[19]:
```python
median_salaries = df.groupby('experience_years')['salary'].median()

distinct_experience = sorted(df['experience_years'].unique())
print("Distinct Experience Years:", distinct_experience)

print("-" *30)

is_consecutive = (median_salaries.index.to_series().diff().dropna() == 1).all()
print(f"Are years consecutive? {is_consecutive}")

print("-" * 30)

print(df['experience_years'].value_counts().sort_index())
```

```
Distinct Experience Years: [np.int64(0), np.int64(1), np.int64(2), np.int64(3),
np.int64(4), np.int64(5), np.int64(6), np.int64(7), np.int64(8), np.int64(9),
np.int64(10), np.int64(11), np.int64(12), np.int64(13), np.int64(14),
np.int64(15), np.int64(16), np.int64(17), np.int64(18), np.int64(19),
np.int64(20)]
------------------------------
Are years consecutive? True
------------------------------
experience_years
0     11804
1     12011
2     11950
3     11890
4     12001
5     11862
6     11860
7     11981
8     11822
9     11856
10    11777
11    11747
12    11944
13    11699
14    12166
15    12052
16    11908
17    11735
18    12102
19    11875
20    11958
Name: count, dtype: int64
```

Before proceeding with the Year-over-Year growth analysis, it is crucial to validate the structure and volume of our data to ensure the mathematical results are both accurate and representative.

1. Why check for Consecutive Years?

```
Mathematical Accuracy: To calculate the Yearly Increase and Growth Rate (%) using methods like

The Logic: If a year is missing (e.g., jumping from 5 to 7 years), the calculation would incor
```

2. Why check Value Counts per Year?

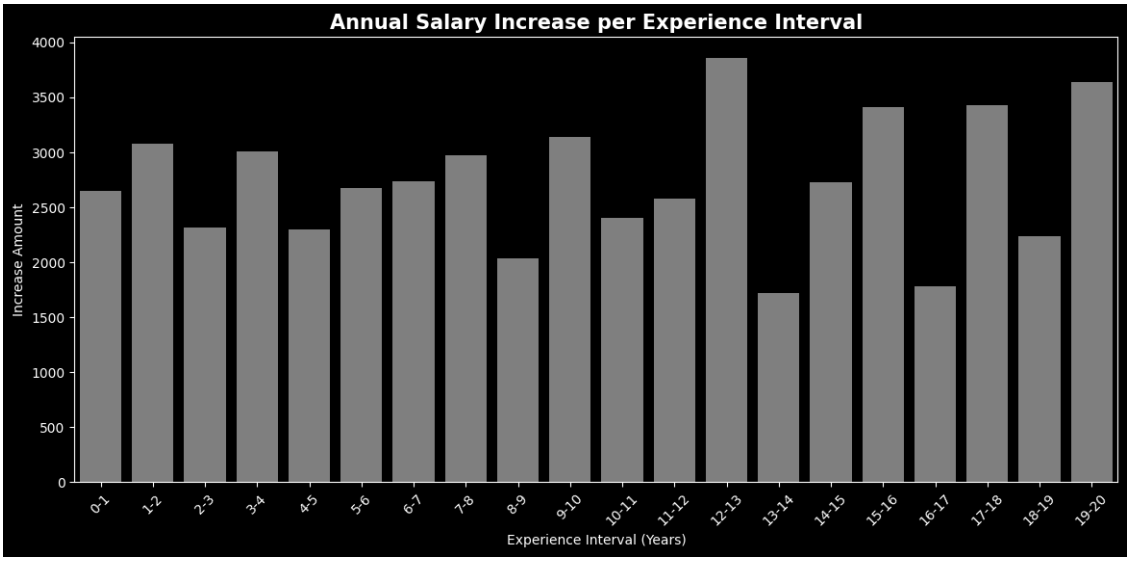
   Statistical Significance: With a massive dataset of 250,000 records, we need to ensure that each "year of experience" category has a sufficient number of observations.

   Reliability of the Median: A median calculated from 10,000 people is much more "stable" and "trustworthy" than a median calculated from only 10 or 20 people. Checking the value_counts() allows us to identify if certain seniority levels have enough data to be considered a reliable market standard.

[20]:
```python
yearly_increase = median_salaries.diff()
percentage_growth = median_salaries.pct_change() * 100
intervals = [f"{int(i-1)}-{int(i)}" for i in median_salaries.index[1:]]

growth_report = pd.DataFrame({
    'Experience Gap': intervals,
    'Annual Increase': yearly_increase.iloc[1:].values,
    'Growth Rate (%)': percentage_growth.iloc[1:].values
})

print(growth_report)
```

```
   Experience Gap  Annual Increase  Growth Rate (%)
0             0-1           2652.5         2.266774
1             1-2           3075.5         2.570006
2             2-3           2319.0         1.889290
3             3-4           3012.5         2.408776
4             4-5           2295.5         1.792295
5             5-6           2676.0         2.052596
6             6-7           2739.5         2.059039
7             7-8           2969.0         2.186513
8             8-9           2038.0         1.468765
9            9-10           3136.0         2.227368
10          10-11           2404.0         1.670256
11          11-12           2581.0         1.763773
12          12-13           3856.0         2.589397
13          13-14           1716.0         1.123250
14          14-15           2727.5         1.765521
15          15-16           3414.5         2.171873
16          16-17           1784.0         1.110634
17          17-18           3426.5         2.109745
18          18-19           2237.5         1.349196
19          19-20           3640.0         2.165674
```

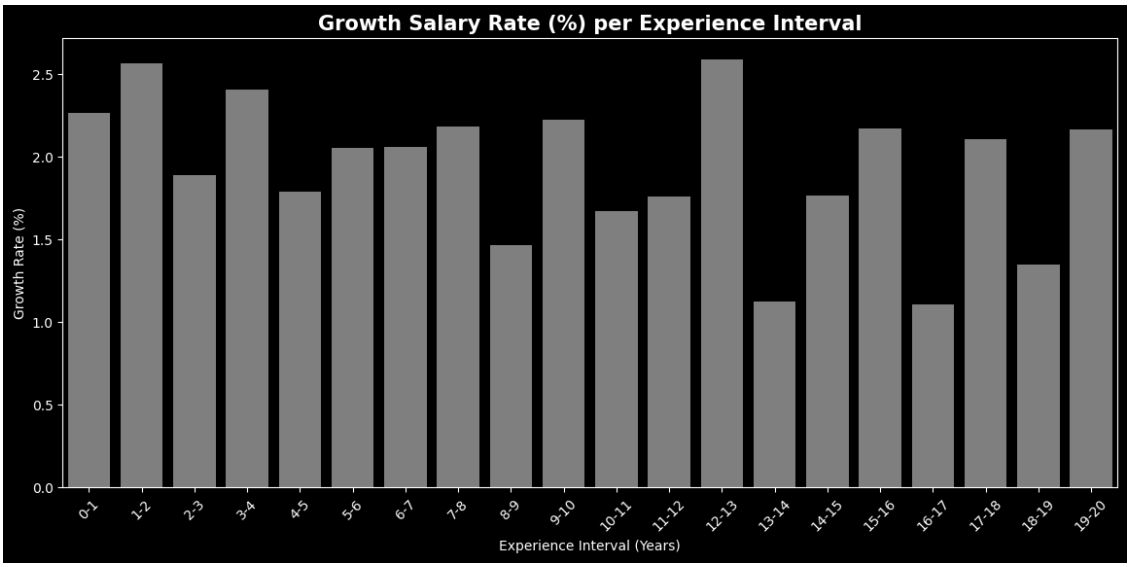
[21]:
```
plt.figure(figsize=(14, 6))

sns.barplot(data=growth_report, x='Experience Gap', y='Annual Increase',␣
↪color="#fff", alpha=0.5)

plt.title('Annual Salary Increase per Experience Interval', fontsize=15,␣
↪fontweight='bold')
plt.xlabel('Experience Interval (Years)')
plt.ylabel('Increase Amount')
plt.xticks(rotation=45)
plt.show()
```



From Annual Salary Increase ($) per Experience Interval bar chart:

Volatility in Growth: The annual salary increase does not follow a predictable linear path, showing significant fluctuation between years.

[22]:
```
plt.figure(figsize=(14, 6))

sns.barplot(data=growth_report, x='Experience Gap', y='Growth Rate␣
↪(%)',color='#fff',alpha=0.5)

plt.title('Growth Salary Rate (%) per Experience Interval', fontsize=15,␣
↪fontweight='bold')
plt.xlabel('Experience Interval (Years)')
plt.ylabel('Growth Rate (%)')
plt.xticks(rotation=45)
plt.show()
```



Volatility in Growth:

The annual salary growth rate does not follow a predictable linear path, showing significant fluctuations across experience intervals.

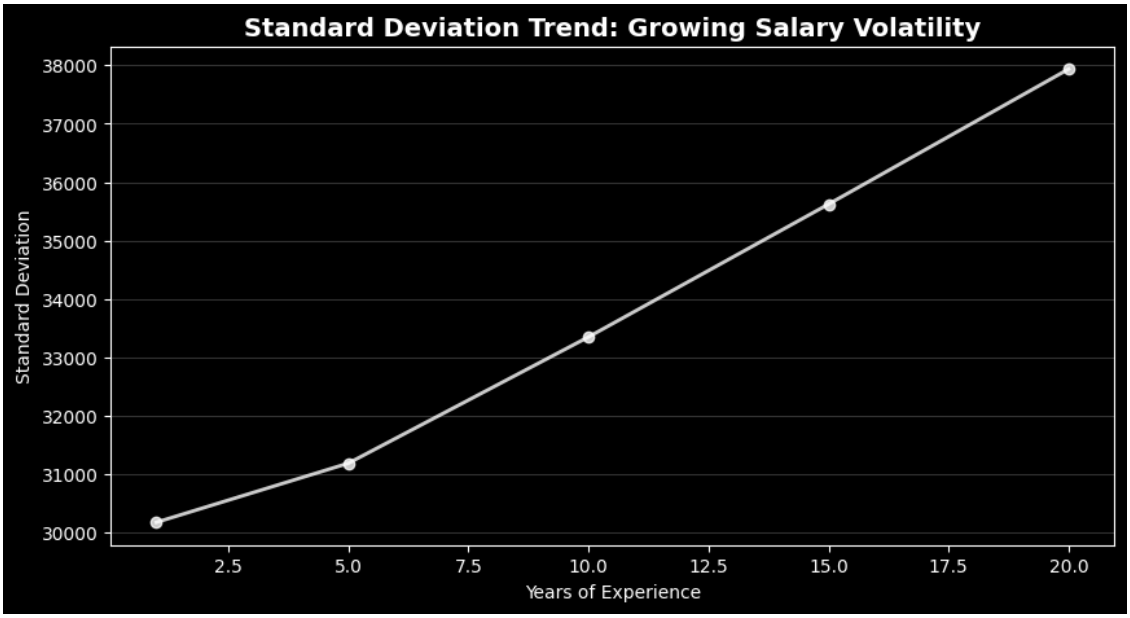
II-1-4. Standard Deviation Trend: Growing Salary Volatility

[23]:
```
std_by_year = df.groupby('experience_years')['salary'].std()

target_years = [1, 5, 10, 15, 20]
target_std = std_by_year.loc[target_years]

std_report = target_std.reset_index()
std_report.columns = ['Years of Experience', 'Standard Deviation']
print(std_report)
```

```
   Years of Experience  Standard Deviation
0                    1        30178.353663
1                    5        31189.351867
2                   10        33349.987461
3                   15        35622.615018
4                   20        37934.873519
```

[24]:
```
plt.figure(figsize=(10, 5))
plt.plot(std_report['Years of Experience'], std_report['Standard Deviation'],
         marker='o', color='#fff', alpha=0.8, linewidth=2)

plt.title('Standard Deviation Trend: Growing Salary Volatility', fontsize=14,␣
 ↪fontweight='bold')
plt.xlabel('Years of Experience')
```

```
plt.ylabel('Standard Deviation')
plt.grid(axis='y', alpha=0.2)
plt.show()
```



Technical Analysis: Standard Deviation as an Indicator of Market Volatility

Linear Development of Risk: Standard deviation follows a consistently upward linear path. This indicates a structural shift in the market as professionals gain more experience.

Prediction vs. Variance: In the early stages of a career (first year), a low standard deviation (around 30,000) suggests a more uniform salary scale compared to those with more years of experience. However, by year 20, the variance widens to approximately $38,000, demonstrating that more years of experience lead to greater salary variance.

Results Interpretation: From Standardization to Divergence

The linear growth of Standard Deviation alongside increasing Salary Variance leads to several key insights:

- As careers mature, Seniority does not guarantee a high salary; rather, it expands the potential range, meaning the gap between top performers and the median wider significantly over time.
- Market Maturity: The rising variance proves that the industry rewards long-term specialization disproportionately. Tenure acts as a "multiplier" for those who successfully position themselves in high-demand sub-sectors, creating the "Outliers" seen in the 300k+ range.

II-1-5. Salary Comparison by Experience and Remote Work Status

[25]:
```
years_to_show = [1, 5, 10, 15, 20]
df_filtered = df[df['experience_years'].isin(years_to_show)]
```

```
plt.figure(figsize=(14, 8))
sns.boxplot(data=df_filtered, x='experience_years', y='salary',␣
 ↪hue='remote_work',
            palette='viridis', showfliers=True)

plt.title('Salary Comparison by Experience and Remote Work Status',␣
 ↪fontsize=15, fontweight='bold')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.legend(title='Remote Work Type')
plt.grid(axis='y', alpha=0.3)
plt.show()
```

Analysis: The Impact of Remote Work on Salary Trajectory

Remote Work Advantage: Across all experience levels, fully remote roles consistently outperfo

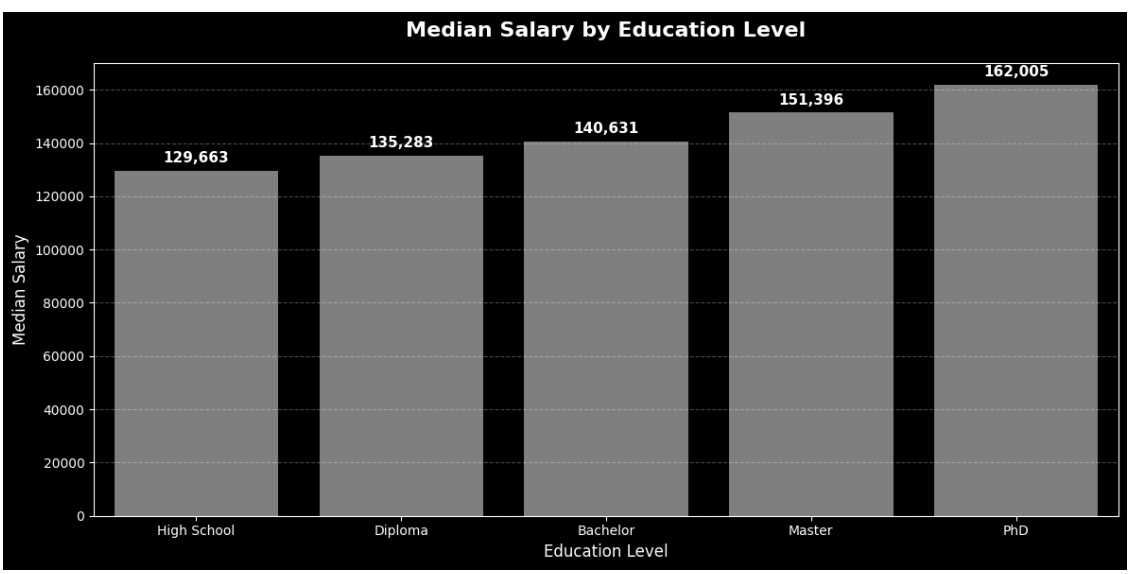
Market Standard: Hybrid and Non-remote roles show similar salary distributions, suggesting tha

1.4.2 II-2. Analyzing the Correlation of Education Level with Salary

II-2-1. Median Salary by Education Level

[49]:
```
education_order = ['High School', 'Diploma', 'Bachelor', 'Master', 'PhD']

plt.figure(figsize=(12, 6))
```

```
sns.barplot(
    data=df,
    x='education_level',
    y='salary',
    order=education_order,
    estimator='median',
    errorbar=None,
    color="#fff",
    alpha=0.5
)

plt.title('Median Salary by Education Level', fontsize=16, fontweight='bold',␣
 ↪pad=20)
plt.xlabel('Education Level', fontsize=12)
plt.ylabel('Median Salary', fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.3)

medians = df.groupby('education_level')['salary'].median().loc[education_order]
for i, value in enumerate(medians):
    plt.text(i, value + 2000, f'{int(value):,}', ha='center', va='bottom',␣
 ↪fontweight='bold', fontsize=11)


plt.tight_layout()
plt.show()
```

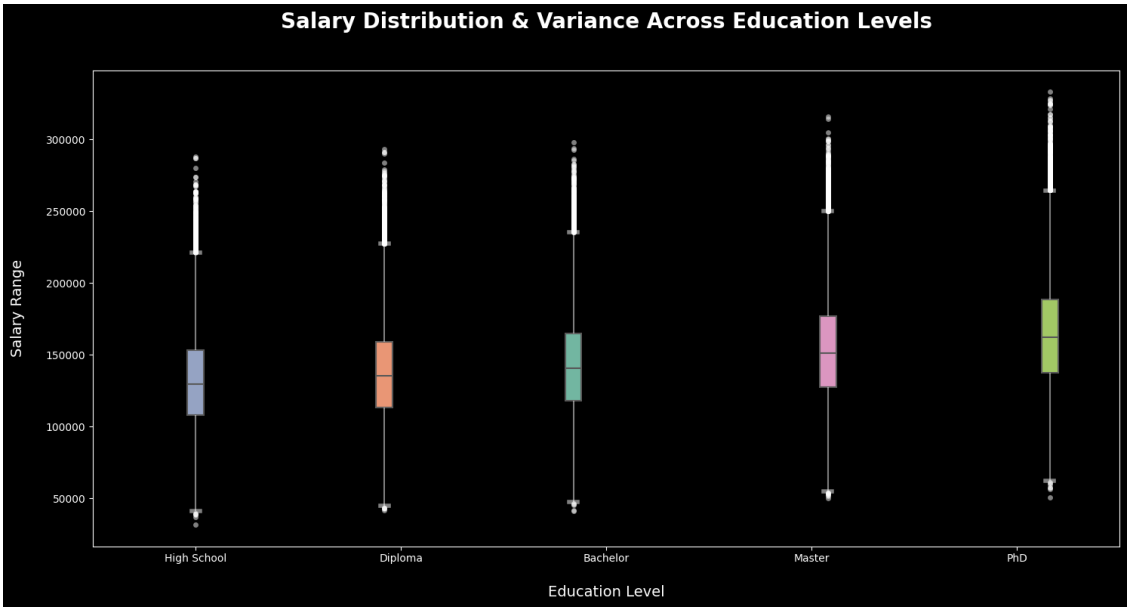


Consistent Growth:

- There is a clear linear progression where higher education levels directly correlate with increased median salaries.
- PhD Dominance: PhD holders command the highest median salary (162,005)
- Educational Floor: The relatively high baseline for High School graduates indicates a lucrative industry

II-2-2. Salary Distribution & Variance Across Education Levels

[27]:
```
plt.style.use('dark_background')

fig, ax = plt.subplots(figsize=(16, 9))

education_order = ['High School', 'Diploma', 'Bachelor', 'Master', 'PhD']

sns.boxplot(
    data=df,
    x='education_level',
    y='salary',
    order=education_order,
    hue='education_level',
    palette='Set2',
    legend=False,
    linewidth=1.5,
    width=0.4,
    ax=ax,
    flierprops={
        'marker': 'o',
        'markersize': 5,
        'markerfacecolor': 'auto',
        'markeredgecolor': 'none',
        'alpha': 0.5
    },
    whiskerprops={'color': '#ffffff', 'linewidth': 1.5, 'alpha': 0.5},
    capprops={'color': '#ffffff', 'linewidth': 4, 'alpha': 0.5}
)

ax.set_title('Salary Distribution & Variance Across Education Levels',
             fontsize=20, fontweight='bold', pad=40)

ax.set_xlabel('Education Level', fontsize=14, labelpad=20)
ax.set_ylabel('Salary Range', fontsize=14, labelpad=20)


plt.subplots_adjust(top=0.85, bottom=0.15, left=0.1, right=0.95)

plt.show()
```



This box chart provides a comprehensive statistical overview of how wages relate to academic qualifications. It reveals the full picture beyond simple averages, focusing on volatility, outliers, and the minimum income threshold.

1. **The "Minimum Income" Effect**

There is a clear direct correlation between educational level and minimum wage.

The bottom lines of the rectangle (representing the lowest 25% of earners) rise steadily from high school to doctorate.

2. **Significant Overlap Between Quartiles**

Despite the high median, there is significant overlap between the boxes (the middle half of the data).

A high school graduate can earn more than someone with a doctorate for several reasons.

Academic qualifications are not the sole determinant of salary. Factors such as specialized skills, years of experience, country, and others also influence pay.

II-2-3. The Interplay between Education & Experience on Compensation

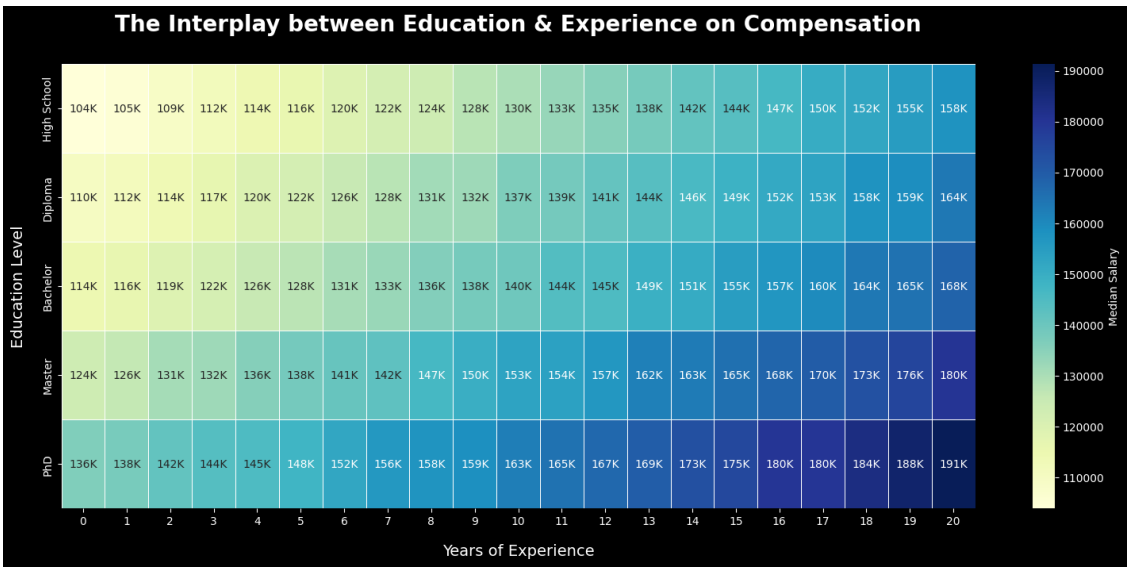
[28]:
```
pivot_df = df.pivot_table(
    index='education_level',
    columns='experience_years',
    values='salary',
    aggfunc='median'
).reindex(['High School', 'Diploma', 'Bachelor', 'Master', 'PhD'])
```

[29]:
```
pivot_df
```

[29]:
```
experience_years        0         1         2         3         4         5   \
education_level
High School        103883.0  105300.5  109081.0  111767.0  114254.0  115870.5
Diploma            110234.0  111650.0  114102.5  116739.5  119975.0  122045.5
Bachelor           114488.0  116437.0  119379.0  121898.5  125598.0  127560.0
Master             123970.0  126304.0  131129.5  131908.0  135940.0  138488.0
PhD                136343.5  138009.0  141528.0  143979.0  145207.0  147817.0

experience_years        6         7         8         9   …        11  \
education_level                                           …
High School        119501.0  122267.0  124018.5  128042.0  …  132917.0
Diploma            125699.0  127996.0  131219.0  132084.5  …  138634.0
Bachelor           130586.0  132607.0  136177.0  138148.0  …  144399.0
Master             140984.0  142215.0  146833.0  149764.5  …  153958.0
PhD                152277.5  156404.0  157563.0  158649.0  …  165224.0

experience_years        12        13        14        15        16        17  \
education_level
High School        135371.0  138250.0  141736.5  143787.0  146681.0  150445.0
Diploma            141191.0  143765.0  146039.5  148615.5  151621.0  153336.0
Bachelor           145329.0  149372.0  150806.0  154802.0  156836.5  159636.5
Master             156622.0  162017.5  163025.0  165446.0  167741.5  169811.0
PhD                167226.0  169449.0  173101.0  175144.5  179785.0  179972.0

experience_years        18        19        20
education_level
High School        152088.5  155379.0  158037.0
Diploma            157897.5  158973.0  163682.5
Bachelor           163772.5  165023.0  168131.0
Master             172587.0  175570.0  180244.0
PhD                183599.0  187581.0  191260.5

[5 rows x 21 columns]
```

[30]:
```python
plt.style.use('dark_background')

annot_text = pivot_df.map(lambda x: f'{x/1000:.0f}K' if pd.notnull(x) else '')

fig, ax = plt.subplots(figsize=(18, 8))

sns.heatmap(
    pivot_df,
    annot=annot_text,
    fmt="",
    cmap="YlGnBu",
    linewidths=.5,
    cbar_kws={'label': 'Median Salary'},
```

```
    ax=ax
)

ax.set_title('The Interplay between Education & Experience on Compensation',
             fontsize=20, fontweight='bold', pad=30)

ax.set_xlabel('Years of Experience', fontsize=14, labelpad=15)
ax.set_ylabel('Education Level', fontsize=14, labelpad=15)

plt.subplots_adjust(top=0.85, bottom=0.15, left=0.15, right=0.95)

plt.show()
```



Experience as a Key Factor

The horizontal color gradient confirms that years of experience consistently contribute to salary growth at all educational levels. Even without a college degree, 20 years of experience can increase a high school graduate's salary by approximately 52% (from $104,000 to $158,000).

The Trade-off Between Degree and Experience

The heat map demonstrates that professional experience effectively bridges the salary gap between academic qualifications and academic qualifications. For example, a professional with a bachelor's degree and 10 years of experience earns $140,000 more than someone with a recent master's degree ($124,000–$126,000). This shows that practical experience is highly valued and can compensate for lower academic qualifications over time.

The "Master's and PhD Advantage"

Advanced degrees offer a "better start" and faster career progression. While it takes a bachelor's degree holder twenty years to earn a salary of $168,000, a PhD holder reaches the same level within

12 years of experience.

1.4.3 II-3. Country-wise Comparison

II-3.1. Statistical Summary: Median Salary by Country

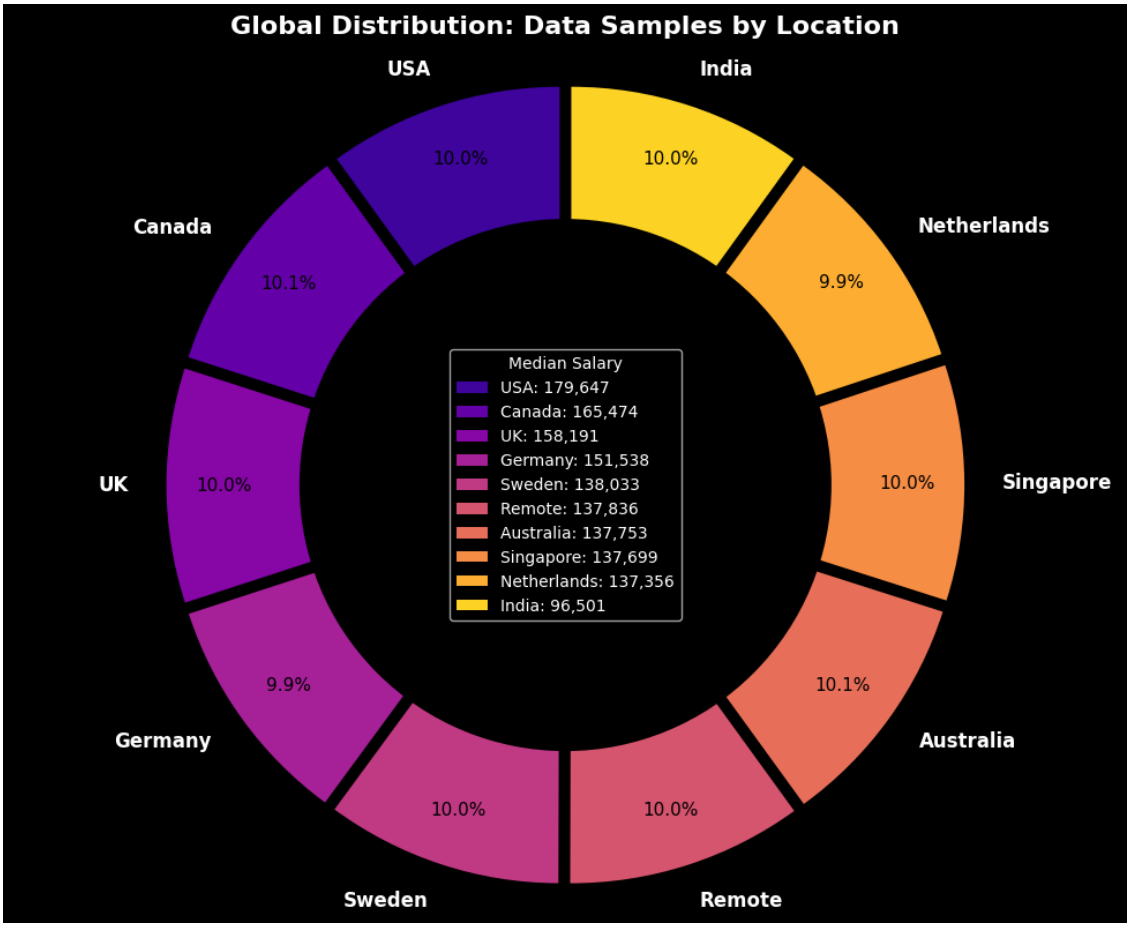
[31]:
```
country_analysis = df.groupby('location')['salary'].agg(['median', 'count']).
 ↪sort_values(by='median', ascending=False)
country_analysis
```

[31]:

	median	count
location		
USA	179647.0	24931
Canada	165474.0	25165
UK	158191.0	24927
Germany	151538.0	24763
Sweden	138033.0	25100
Remote	137836.0	25065
Australia	137753.5	25258
Singapore	137699.0	25035
Netherlands	137356.0	24861
India	96501.0	24895

[32]:
```
plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(10, 8))

wedges, texts, autotexts = ax.pie(country_analysis['count'],
                                  labels=country_analysis.index,
                                  autopct='%1.1f%%',
                                  startangle=90,
                                  colors=sns.color_palette("plasma",␣
 ↪len(country_analysis)),
                                  pctdistance=0.85,
                                  explode=[0.05] * len(country_analysis))

centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=12, weight="bold", color="white")
plt.setp(autotexts, size=11, color="black")

ax.axis('equal')
plt.title('Global Distribution: Data Samples by Location', fontsize=16,␣
 ↪fontweight='bold', pad=20)

salary_info = [f"{loc}: {int(med):,}" for loc, med in zip(country_analysis.
 ↪index, country_analysis['median'])]
ax.legend(wedges, salary_info, title="Median Salary", loc="center", fontsize=10)
```

```
plt.tight_layout()
plt.show()
```



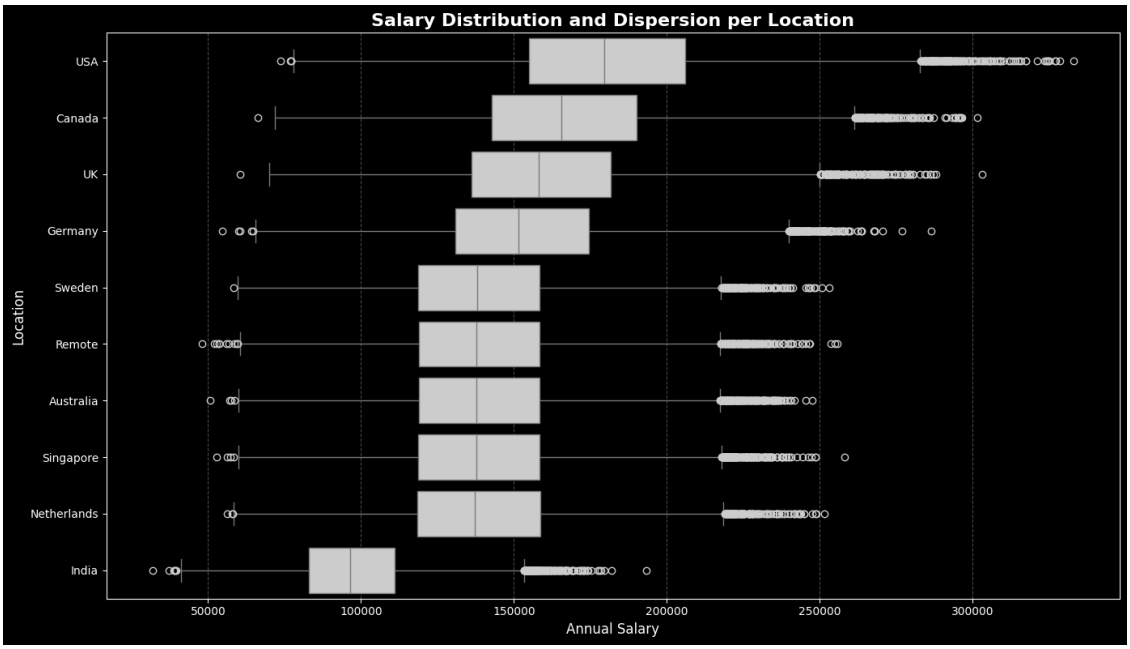
A homogeneous distribution: The graph shows a nearly uniform distribution of data samples across all locations (approximately 10% per location).

Salary disparities: Despite the equal sample sizes, average salaries vary significantly, with the USA topping the list at 179,647 and India at the bottom at 96,501.

II-3-2. Salary Dispersion: Identifying Income Gaps by Location

[33]:
```
plt.figure(figsize=(14, 8))
plt.style.use('dark_background')

sns.boxplot(x='salary', y='location', data=df,
            order=country_analysis.index,
            flierprops={"marker": "o", "markeredgecolor": "#ccc"},
           color="#ccc"            )
```

```
plt.title('Salary Distribution and Dispersion per Location', fontsize=16,␣
↪fontweight='bold')
plt.xlabel('Annual Salary', fontsize=12)
plt.ylabel('Location', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.3)

plt.tight_layout()
plt.show()
```



[50]:
```
stats_table = df.groupby('location')['salary'].agg([
    ('Median', 'median'),
    ('25%', lambda x: x.quantile(0.25)),
    ('75%', lambda x: x.quantile(0.75))
])

stats_table['IQR (Difference)'] = stats_table['75%'] - stats_table['25%']

stats_table = stats_table.sort_values(by='Median', ascending=False)

styled_stats = stats_table.style.format("{:,.0f}")
styled_stats
```

[50]: <pandas.io.formats.style.Styler at 0x7dfb44f20200>

Detailed Notes:

Mechanical Correlation Between Ranges: There is an unusual correlation between the minimum,

maximum, and median values. In real-world datasets, a country with a low median might have very high wages (e.g., few high-income earners), but here, no region with a low median exceeds the maximum wage in countries with the highest median. This suggests that the data may be artificially generated and not based on a naturally occurring random sample.

Mirror Effect (Artificial Symmetry): There is a near-perfect and suspiciously similar match between the samples from Sweden, Australia, Singapore, the Outback, and the Netherlands. These four categories are almost identical across five different statistical indicators:

- Minimum and maximum values.
- Meaning.
- 25th and 75th percentiles (quartiles).

This level of mathematical symmetry across diverse geographical locations strongly supports the hypothesis that this dataset may be artificial.

Interquartile Range (IQR) Analysis: * USA: Shows the largest difference between the 75th and 25th percentiles (interquartile range), indicating the most flexible salary distribution in this dataset.

India: Shows the smallest difference (interquartile range), indicating a very narrow salary range that lacks the diversity typical of a real labor market.

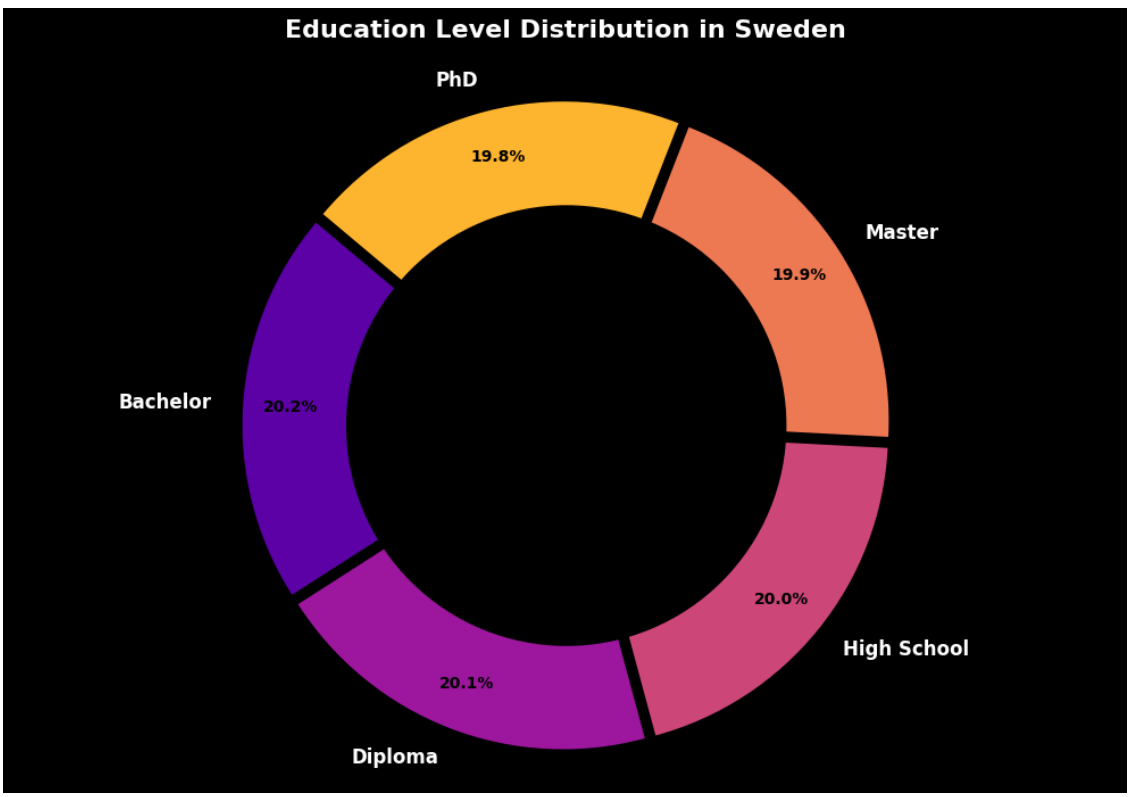
Media-Dispersion Relationship: We observe a direct, positive relationship between the median and the interquartile range. As the median increases, the gap between the 75th and 25th percentiles increases proportionally.

II-3-3. Data Samples by Education Level

[35]:
```
edu_loc_table = pd.crosstab(df['location'], df['education_level'])
display(edu_loc_table)

sweden_data = edu_loc_table.loc['Sweden']

plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(10, 7))

wedges, texts, autotexts = ax.pie(sweden_data,
                                  labels=sweden_data.index,
                                  autopct='%1.1f%%',
                                  startangle=140,
                                  colors=sns.color_palette("plasma",␣
 ↪len(sweden_data)),
                                  pctdistance=0.85,
                                  explode=[0.03] * len(sweden_data))

centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=12, weight="bold")
plt.setp(autotexts, size=10, color="black", weight="bold")
```

```
plt.title('Education Level Distribution in Sweden', fontsize=16,␣
 ↪fontweight='bold', pad=20)
ax.axis('equal')
plt.tight_layout()
plt.show()
```

```
education_level  Bachelor  Diploma  High School  Master   PhD
location
Australia            4983     4997         5169    5102  5007
Canada               5025     4997         4952    5160  5031
Germany              4868     4927         5053    4958  4957
India                5109     4922         4932    4956  4976
Netherlands          4912     4922         5041    5009  4977
Remote               4990     4973         4993    5095  5014
Singapore            5069     5053         4935    4996  4982
Sweden               5065     5052         5013    4995  4975
UK                   4997     5010         4990    4974  4956
USA                  4932     4923         4987    5107  4982
```



Observation and Analysis:

The "Constant Number" Phenomenon: When examining the distribution of educational levels in

different geographical locations (such as Sweden, the USA, and India), we observe a remarkable similarity in the number of samples for each category.

The 20% Uniformity Rule: In each country, each educational level (high school, bachelor's degree, etc.) represents approximately 20% of the total sample.

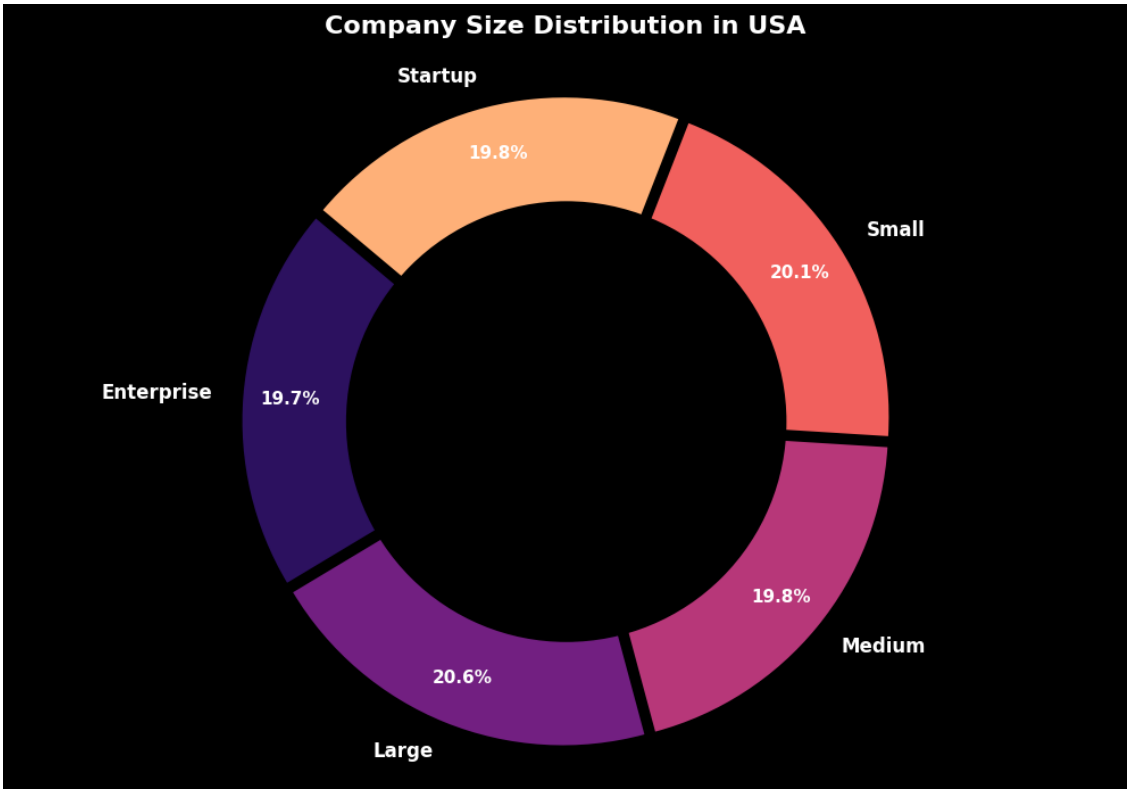
This near-perfect similarity strongly supports the hypothesis of artificial data.

II-3-4. Corporate Landscape: Company Size Distribution by Location

[36]:
```
company_loc_table = pd.crosstab(df['location'], df['company_size'])
display(company_loc_table)

usa_company_data = company_loc_table.loc['USA']

plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(10, 7))

wedges, texts, autotexts = ax.pie(usa_company_data,
                                  labels=usa_company_data.index,
                                  autopct='%1.1f%%',
                                  startangle=140,
                                  colors=sns.color_palette("magma",␣
 ↪len(usa_company_data)),
                                  pctdistance=0.85,
                                  explode=[0.03] * len(usa_company_data))

centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=12, weight="bold")
plt.setp(autotexts, size=11, color="white", weight="bold")

plt.title('Company Size Distribution in USA', fontsize=16, fontweight='bold',␣
 ↪pad=20)
ax.axis('equal')
plt.tight_layout()
plt.show()
```

company_size	Enterprise	Large	Medium	Small	Startup
location					
Australia	4988	5050	5108	5135	4977
Canada	5016	4913	5058	5055	5123
Germany	4850	5092	4943	4993	4885
India	5002	4975	4985	4934	4999
Netherlands	5027	4967	5071	4920	4876
Remote	5059	5044	4832	5084	5046
Singapore	4966	5024	5007	5101	4937
Sweden	5080	5085	5037	4948	4950
UK	4973	4973	5043	5061	4877

```
USA                    4914    5131     4943    5004      4939
```



Observation and Analysis:

The **"Constant Number"** Phenomenon: When examining the distribution of company sizes in different geographical locations (such as Sweden, the USA, and India), we observe a remarkable similarity in the number of samples for each category.

The 20% Uniformity Rule: In each country, each company size represents approximately 20% of the total sample.

Conclusion: This near-perfect similarity strongly supports the hypothesis of artificial data.

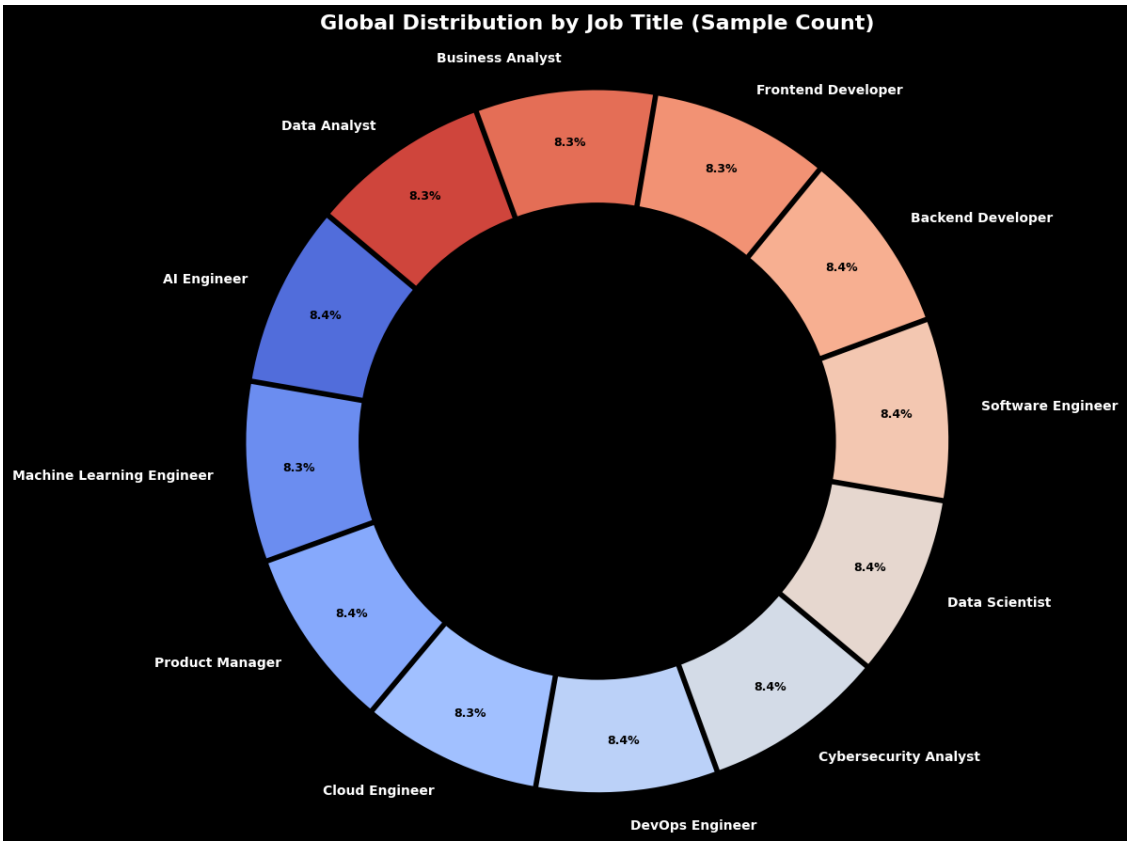
1.4.4 II-4. Job title comparison

II-4-1. Statistical Summary: Median Salary by Job Title

[37]:
```
job_stats = df.groupby('job_title')['salary'].agg(['median', 'count']).
 ↪sort_values(by='median', ascending=False)
display(job_stats)

plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(12, 9))

wedges, texts, autotexts = ax.pie(job_stats['count'],
```

```
                                    labels=job_stats.index,
                                    autopct='%1.1f%%',
                                    startangle=140,
                                    colors=sns.color_palette("coolwarm",␣
 ↪len(job_stats)),
                                    pctdistance=0.85,
                                    explode=[0.03] * len(job_stats))

centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=10, weight="bold")
plt.setp(autotexts, size=9, color="black", weight="bold")

plt.title('Global Distribution by Job Title (Sample Count)', fontsize=16,␣
 ↪fontweight='bold', pad=20)
ax.axis('equal')
plt.tight_layout()
plt.show()
```

```
                            median  count
job_title
AI Engineer                171674.0  20945
Machine Learning Engineer  160933.0  20677
Product Manager            155655.5  20950
Cloud Engineer             150005.0  20686
DevOps Engineer            148021.0  20889
Cybersecurity Analyst      146737.0  20959
Data Scientist             145231.5  20890
Software Engineer          139432.5  20876
Backend Developer          136678.0  21125
Frontend Developer         130619.0  20633
Business Analyst           120562.0  20648
Data Analyst               118079.5  20722
```



The "Constant Number" Phenomenon: When examining the distribution of the 12 main job titles (such as Data Scientist, ML Engineer, Data Analyst, etc.), we observe a remarkable similarity in the number of samples for each category.

The 8.3% Uniformity Rule: Within the dataset, each of the 12 job titles represents approximately 8.3% of the total sample (100/12 8.3%). In a real-world labor market.

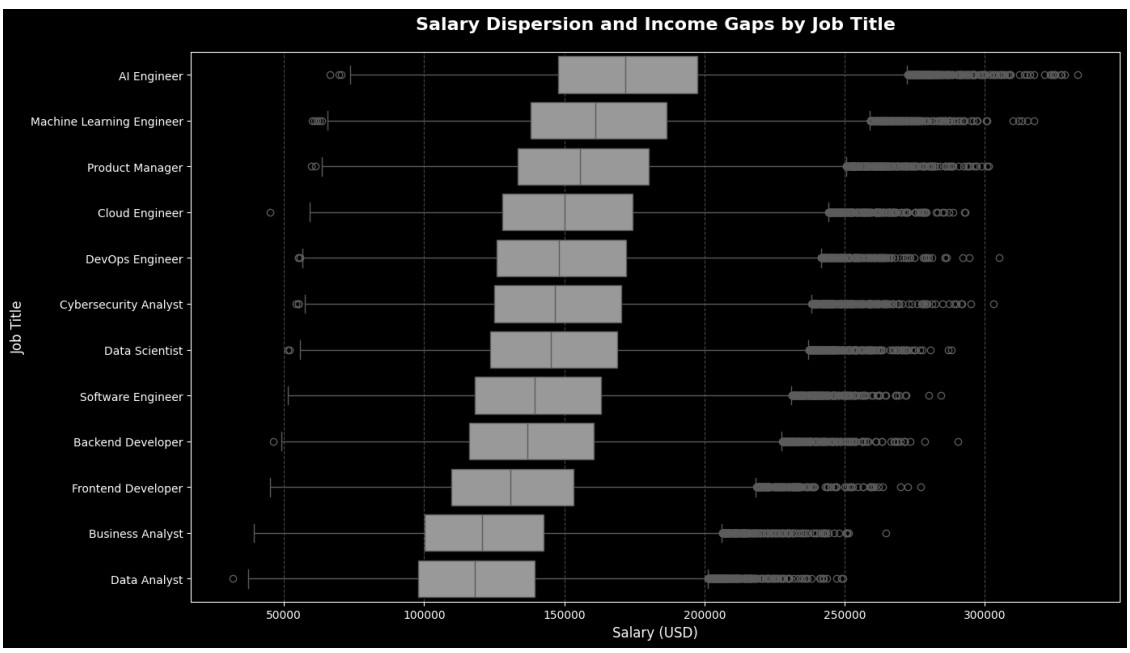
Conclusion: This near-perfect similarity across 12 distinct categories strongly supports the hypothesis of artificial data.

II-4-2. Salary Dispersion: Identifying Income Gaps by Job

[38]:
```
job_dispersion = df.groupby('job_title')['salary'].agg(['min', 'max', 'median',␣
↪lambda x: x.quantile(0.75) - x.quantile(0.25)]).rename(columns={'<lambda_0>':
↪ 'IQR'}).sort_values(by='median', ascending=False)
display(job_dispersion)

plt.style.use('dark_background')
plt.figure(figsize=(14, 8))

sns.boxplot(x='salary', y='job_title', data=df, order=job_dispersion.
↪index,color="#999")
```

```
plt.title('Salary Dispersion and Income Gaps by Job Title', fontsize=16,␣
 ↪fontweight='bold', pad=20)
plt.xlabel('Salary (USD)', fontsize=12)
plt.ylabel('Job Title', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.3)
plt.tight_layout()
plt.show()
```

job_title	min	max	median	IQR
AI Engineer	66371	333046	171674.0	49835.00
Machine Learning Engineer	59925	317377	160933.0	48416.00
Product Manager	59796	301294	155655.5	46941.00
Cloud Engineer	45030	292855	150005.0	46491.00
DevOps Engineer	54985	304968	148021.0	46239.00
Cybersecurity Analyst	54224	303178	146737.0	45272.50
Data Scientist	51349	288049	145231.5	45421.50
Software Engineer	51313	284442	139432.5	45240.25
Backend Developer	46304	290512	136678.0	44613.00
Frontend Developer	44964	277053	130619.0	43506.00
Business Analyst	39285	264567	120562.0	42368.00
Data Analyst	31867	249354	118079.5	41317.50



[39]:
```
job_salary_stats = df.groupby('job_title')['salary'].agg([
    ('Median', 'median'),
    ('25%', lambda x: x.quantile(0.25)),
```

```
    ('75%', lambda x: x.quantile(0.75)),
    ('IQR (75%-25%)', lambda x: x.quantile(0.75) - x.quantile(0.25))
]).sort_values(by='Median', ascending=False)

display(job_salary_stats)
```

job_title	Median	25%	75%	IQR (75%-25%)
AI Engineer	171674.0	147656.00	197491.00	49835.00
Machine Learning Engineer	160933.0	138044.00	186460.00	48416.00
Product Manager	155655.5	133200.75	180141.75	46941.00
Cloud Engineer	150005.0	127965.00	174456.00	46491.00
DevOps Engineer	148021.0	125860.00	172099.00	46239.00
Cybersecurity Analyst	146737.0	124981.00	170253.50	45272.50
Data Scientist	145231.5	123540.75	168962.25	45421.50
Software Engineer	139432.5	117949.75	163190.00	45240.25
Backend Developer	136678.0	115936.00	160549.00	44613.00
Frontend Developer	130619.0	109773.00	153279.00	43506.00
Business Analyst	120562.0	100276.50	142644.50	42368.00
Data Analyst	118079.5	97952.50	139270.00	41317.50

Observation and Analysis:

Linear Gradient: We observe a consistent and clear correlation, where the minimum and maximum salaries increase proportionally with the median. In a typical job market, some jobs (such as data analyst) may have a high maximum salary due to the presence of experienced candidates, even if the median is low. However, in this case, the entire range moves as a single, fixed unit.

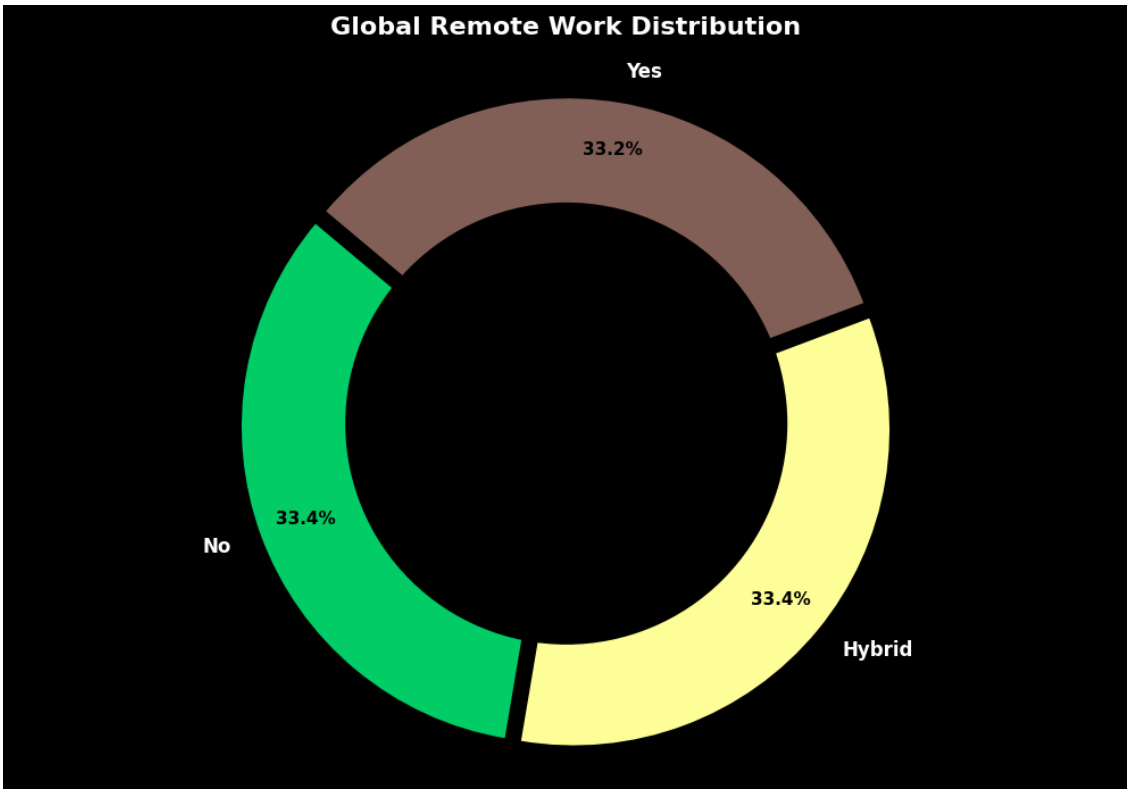
Relative Dispersion (Interquartile Range): There is a clear and consistent relationship between the median and the interquartile range (75% - 25%). As the average salary for a given job increases, the gap between the 25th and 75th percentiles widens.

II-4-3. Remote Work: Distribution Across Job Titles

[40]:
```
remote_job_table = pd.crosstab(df['job_title'], df['remote_work'])
display(remote_job_table)

remote_global_dist = df['remote_work'].value_counts()

plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(10, 7))

wedges, texts, autotexts = ax.pie(remote_global_dist,
                                  labels=remote_global_dist.index,
                                  autopct='%1.1f%%',
                                  startangle=140,
                                  colors=sns.color_palette("terrain",␣
 ↪len(remote_global_dist)),
                                  pctdistance=0.85,
                                  explode=[0.03] * len(remote_global_dist))
```

```
centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=12, weight="bold")
plt.setp(autotexts, size=11, color="black", weight="bold")

plt.title('Global Remote Work Distribution', fontsize=16, fontweight='bold',␣
 ↪pad=20)
ax.axis('equal')
plt.tight_layout()
plt.show()
```

```
remote_work                Hybrid    No   Yes
job_title
AI Engineer                  7049  6902  6994
Backend Developer            7139  6965  7021
Business Analyst             6916  6883  6849
Cloud Engineer               6948  6896  6842
Cybersecurity Analyst        6936  7044  6979
Data Analyst                 6829  6987  6906
Data Scientist               7074  6906  6910
DevOps Engineer              6919  7049  6921
Frontend Developer           6880  6922  6831
Machine Learning Engineer    6906  7005  6766
Product Manager              6883  7156  6911
Software Engineer            6997  6906  6973
```



Observation & Analysis:

The "Constant Number" Phenomenon: When examining the distribution of remote work scenarios (e.g., remote, on-site, hybrid) across the twelve job titles, we observe a remarkable similarity in the number of samples across the entire dataset and also within each job title.

The 33% Rule: Within each job title, each work style represents approximately 1/3 of the samples.

Conclusion: This near-perfect 1/3 distribution across all twelve job titles strongly supports the hypothesis of artificial data.

1.5 III. Is the data synthetic?

[41]:
```
%%html
<section id="synthetic-data-analysis" style="font-family: 'Segoe UI', Tahoma,␣
↪Geneva, Verdana, sans-serif; line-height: 1.6; color: #e0e0e0;␣
↪background-color: #121212; padding: 20px; border-radius: 8px;">

    <h2 style="color: #4db8ff; border-bottom: 2px solid #4db8ff; padding-bottom:
↪ 10px; display: flex; align-items: center;">
        <span style="margin-right: 10px;"> </span> Itroduction
    </h2>
```

```
    <div class="english-text" style="margin-top: 20px; background-color:␣
↪#1e1e1e; padding: 15px; border-left: 5px solid #4db8ff;">
        <p style="font-size: 1.1em; margin: 0;">
            Previously, we presented several pieces of evidence suggesting that␣
↪the data might be artificial. However, we will now dedicate a specific␣
↪section to discuss this topic in detail, using <strong>Job Titles</strong>␣
↪as a  case study to our analysis.
        </p>
    </div>




    </div>

</section>
```

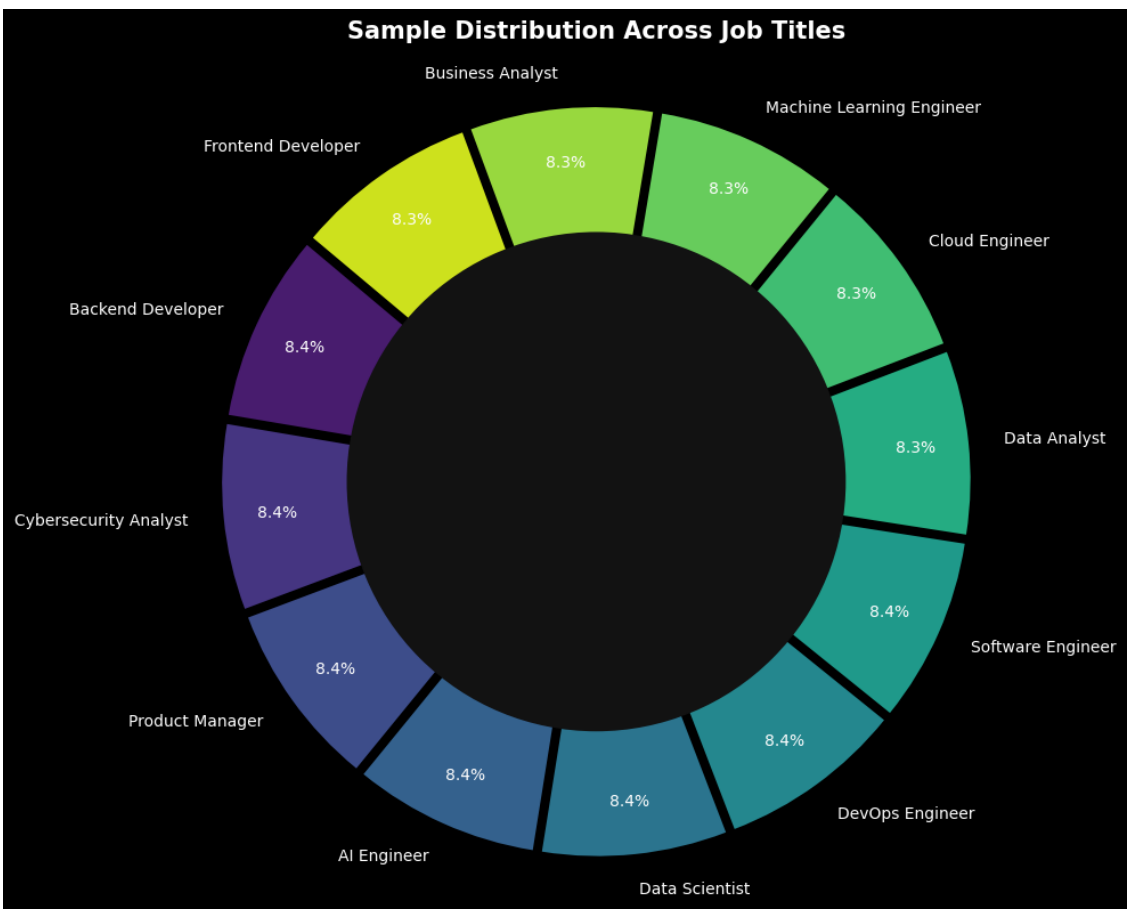
```
<IPython.core.display.HTML object>
```

1.5.1 III-1. Statistical Distribution of Sample Sizes Across Job Titles

[42]:
```
job_counts = df['job_title'].value_counts().reset_index()
job_counts.columns = ['Job Title', 'Sample Count']
job_counts['Percentage (%)'] = (job_counts['Sample Count'] / len(df) * 100).
↪round(2)

print("Job Title Sample Distribution Table:")
display(job_counts)

plt.style.use('dark_background')
plt.figure(figsize=(10, 8))

colors = sns.color_palette("viridis", len(job_counts))

plt.pie(job_counts['Sample Count'],
        labels=job_counts['Job Title'],
        autopct='%1.1f%%',
        startangle=140,
        colors=colors,
        pctdistance=0.85,
        explode=[0.05]*len(job_counts))

centre_circle = plt.Circle((0,0), 0.70, fc='#121212')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

plt.title('Sample Distribution Across Job Titles', fontsize=15,␣
↪fontweight='bold', pad=20)
```

```
plt.axis('equal')
plt.tight_layout()
plt.show()
```

```
Job Title Sample Distribution Table:

                    Job Title  Sample Count  Percentage (%)
0           Backend Developer         21125            8.45
1       Cybersecurity Analyst         20959            8.38
2             Product Manager         20950            8.38
3                 AI Engineer         20945            8.38
4              Data Scientist         20890            8.36
5             DevOps Engineer         20889            8.36
6           Software Engineer         20876            8.35
7               Data Analyst         20722            8.29
8              Cloud Engineer         20686            8.27
9   Machine Learning Engineer         20677            8.27
10           Business Analyst         20648            8.26
11          Frontend Developer         20633            8.25
```



Observation and Analysis:

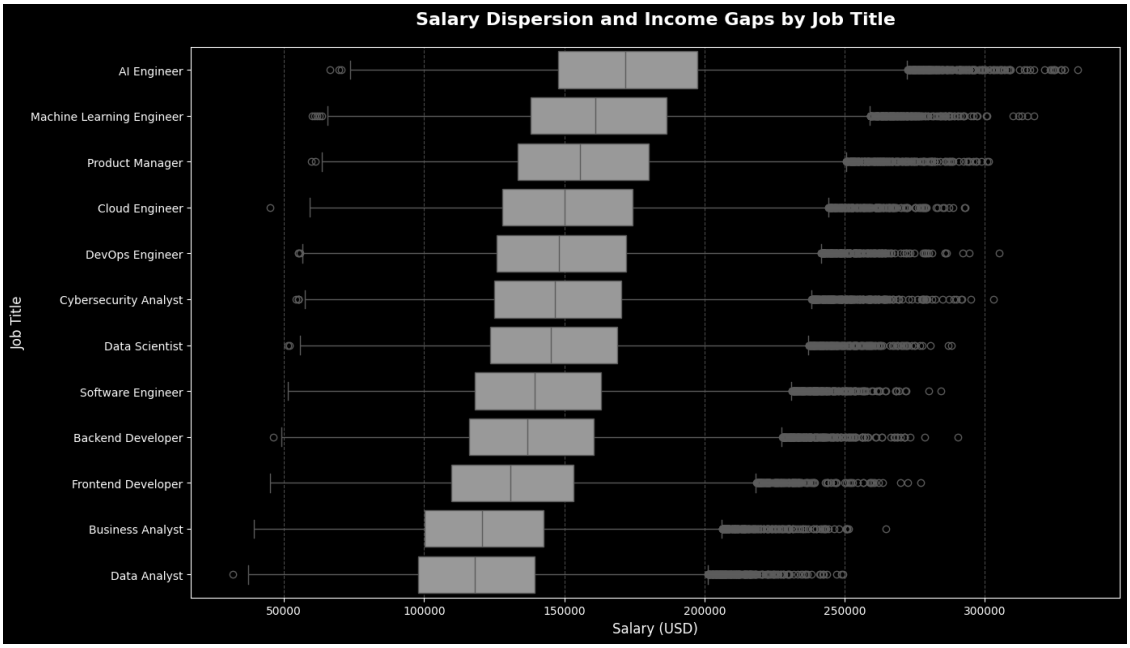
Upon examining the distribution of the dataset, we observe that the number of samples for each of the twelve job titles is remarkably consistent, with each title representing approximately 8.3% of the total data.

Although such a high degree of consistency often suggests artificial data, it is important to consider an alternative systematic interpretation. This homogeneity may also be the result of a controlled data collection strategy, where an equal number of observations were intentionally collected for each job category to ensure balanced representation.

1.5.2 III-2. Salary Dispersion: Identifying Income Gaps by Job

[43]:
```
job_dispersion = df.groupby('job_title')['salary'].agg(['min', 'max', 'median',␣
↪lambda x: x.quantile(0.75) - x.quantile(0.25)]).rename(columns={'<lambda_0>':
↪ 'IQR'}).sort_values(by='median', ascending=False)
display(job_dispersion)

plt.style.use('dark_background')
plt.figure(figsize=(14, 8))

sns.boxplot(x='salary', y='job_title', data=df, order=job_dispersion.
↪index,color="#999")

plt.title('Salary Dispersion and Income Gaps by Job Title', fontsize=16,␣
↪fontweight='bold', pad=20)
plt.xlabel('Salary (USD)', fontsize=12)
plt.ylabel('Job Title', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.3)
plt.tight_layout()
plt.show()
```

```
                            min     max    median       IQR
job_title
AI Engineer               66371  333046  171674.0  49835.00
Machine Learning Engineer 59925  317377  160933.0  48416.00
Product Manager           59796  301294  155655.5  46941.00
Cloud Engineer            45030  292855  150005.0  46491.00
DevOps Engineer           54985  304968  148021.0  46239.00
Cybersecurity Analyst     54224  303178  146737.0  45272.50
Data Scientist            51349  288049  145231.5  45421.50
Software Engineer         51313  284442  139432.5  45240.25
Backend Developer         46304  290512  136678.0  44613.00
Frontend Developer        44964  277053  130619.0  43506.00
Business Analyst          39285  264567  120562.0  42368.00
Data Analyst              31867  249354  118079.5  41317.50
```



[44]:
```
job_salary_stats = df.groupby('job_title')['salary'].agg([
    ('Median', 'median'),
    ('25%', lambda x: x.quantile(0.25)),
    ('75%', lambda x: x.quantile(0.75)),
    ('IQR (75%-25%)', lambda x: x.quantile(0.75) - x.quantile(0.25))
]).sort_values(by='Median', ascending=False)

display(job_salary_stats)
```

job_title	Median	25%	75%	IQR (75%-25%)
AI Engineer	171674.0	147656.00	197491.00	49835.00
Machine Learning Engineer	160933.0	138044.00	186460.00	48416.00
Product Manager	155655.5	133200.75	180141.75	46941.00
Cloud Engineer	150005.0	127965.00	174456.00	46491.00
DevOps Engineer	148021.0	125860.00	172099.00	46239.00
Cybersecurity Analyst	146737.0	124981.00	170253.50	45272.50
Data Scientist	145231.5	123540.75	168962.25	45421.50
Software Engineer	139432.5	117949.75	163190.00	45240.25
Backend Developer	136678.0	115936.00	160549.00	44613.00
Frontend Developer	130619.0	109773.00	153279.00	43506.00
Business Analyst	120562.0	100276.50	142644.50	42368.00
Data Analyst	118079.5	97952.50	139270.00	41317.50

Observation and Analysis:

Linear Gradient: We observe a consistent and clear correlation, where the minimum and maximum salaries increase proportionally with the median. In a typical job market, some jobs (such as data

analyst) may have a high maximum salary due to the presence of experienced candidates, even if the median is low. However, in this case, the entire range moves as a single, fixed unit.

Relative Dispersion (Interquartile Range): There is a clear and consistent relationship between the median and the interquartile range (75% - 25%). As the average salary for a given job increases, the gap between the 25th and 75th percentiles widens. This strongly supports the argument that the data may be artificial.

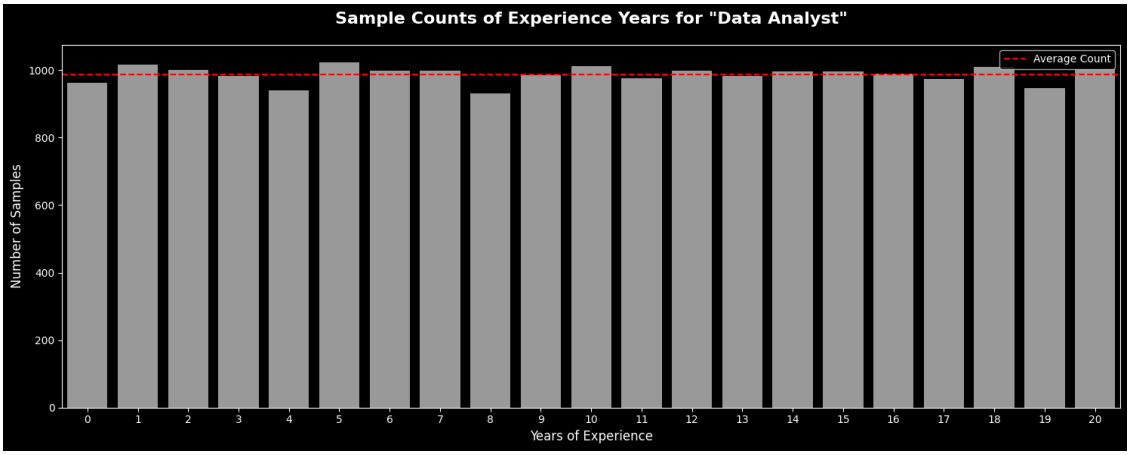
1.5.3 III-3. Experience Years Distribution Across Job Titles: Analyzing Numerical Symmetry

[45]:
```
experience_pivot = pd.crosstab(df['job_title'], df['experience_years'])

all_years = list(range(1, 21))
experience_pivot = experience_pivot.reindex(columns=all_years, fill_value=0)

print("Sample Counts of Experience Years per Job Title (Full Range):")
display(experience_pivot)
```

```
Sample Counts of Experience Years per Job Title (Full Range):

experience_years             1     2     3     4     5     6     7     8   \
job_title
AI Engineer                1003  1041   984  1033   972   977   980  1001
Backend Developer           964  1058  1001  1015   957   975  1027  1012
Business Analyst            996   962   957  1003   987   962  1022   958
Cloud Engineer             1003   956   988  1034   951   968  1030   984
Cybersecurity Analyst      1033   969   986  1026   993   976   992  1024
Data Analyst               1017  1000   982   940  1024   999   999   931
Data Scientist              970   954   957   964   997  1005  1023   983
DevOps Engineer             989  1026  1007   953   979   996  1024  1004
Frontend Developer         1074  1004   998  1033  1002  1021   935   971
Machine Learning Engineer  1075   984   995   971   991   991   977   938
Product Manager             993   978  1033   997   993   974   987  1015
Software Engineer           894  1018  1002  1032  1016  1016   985  1001

experience_years             9    10    11    12    13    14    15    16  \
job_title
AI Engineer                 994  1004   987  1029   979  1006   998   954
Backend Developer          1037  1020   979  1038   984   989  1002  1007
Business Analyst           1045   942   942  1004   950  1030  1010   969
Cloud Engineer              919   985   960  1004  1023  1025   968   989
Cybersecurity Analyst      1010   981  1010  1002   982  1028   975  1031
Data Analyst                984  1011   976   999   982   997   997   990
Data Scientist              952  1005  1038  1014   976  1002   999  1001
DevOps Engineer             976   984   967   976   971  1070  1021  1011
Frontend Developer         1022   973   953   894   930   960  1048   950
Machine Learning Engineer   950   968   966   946   936  1006  1038   981
Product Manager             977   959   982  1025  1027  1055  1017   998
```

```
Software Engineer           990   945   987  1013   959   998   979  1027

experience_years             17    18    19    20
job_title
AI Engineer                1014  1031   995   985
Backend Developer          1039   974  1039   972
Business Analyst            966  1054   995   975
Cloud Engineer              966  1020   996   963
Cybersecurity Analyst       974  1006   978   989
Data Analyst                973  1010   946  1003
Data Scientist             1007  1043  1032   995
DevOps Engineer             919  1024   977  1016
Frontend Developer          937   961   979   992
Machine Learning Engineer   960   968   997  1037
Product Manager            1024   979   918  1037
Software Engineer           956  1032  1023   994
```

[46]:
```
data_analyst_data = df[df['job_title'] == 'Data Analyst']

da_exp_counts = data_analyst_data['experience_years'].value_counts().
 ↪sort_index().reset_index()
da_exp_counts.columns = ['Experience Years', 'Sample Count']

plt.style.use('dark_background')
plt.figure(figsize=(15, 6))

sns.barplot(x='Experience Years', y='Sample Count', data=da_exp_counts,␣
 ↪color='#999')

plt.axhline(da_exp_counts['Sample Count'].mean(), color='red', linestyle='--',␣
 ↪label='Average Count')

plt.title('Sample Counts of Experience Years for "Data Analyst"', fontsize=16,␣
 ↪fontweight='bold', pad=20)
plt.xlabel('Years of Experience', fontsize=12)
plt.ylabel('Number of Samples', fontsize=12)
plt.xticks(rotation=0)
plt.legend()

plt.tight_layout()
plt.show()
```



Result:

A striking and unnatural symmetry is observed in the sample distribution across the dataset. Not only is the number of samples identical for each year of experience (from 1 to 20), but this exact pattern is repeated across all job titles.

Comparison: For example, the experience distribution for a data analyst is a perfect mirror image of the experience distribution for a software engineer. In both cases, the same frequency is allocated to each year of experience.

This strongly supports the argument that this data may be artificially generated.

1.5.4 III-4. Certification Distribution: Cross-Job Symmetry Analysis

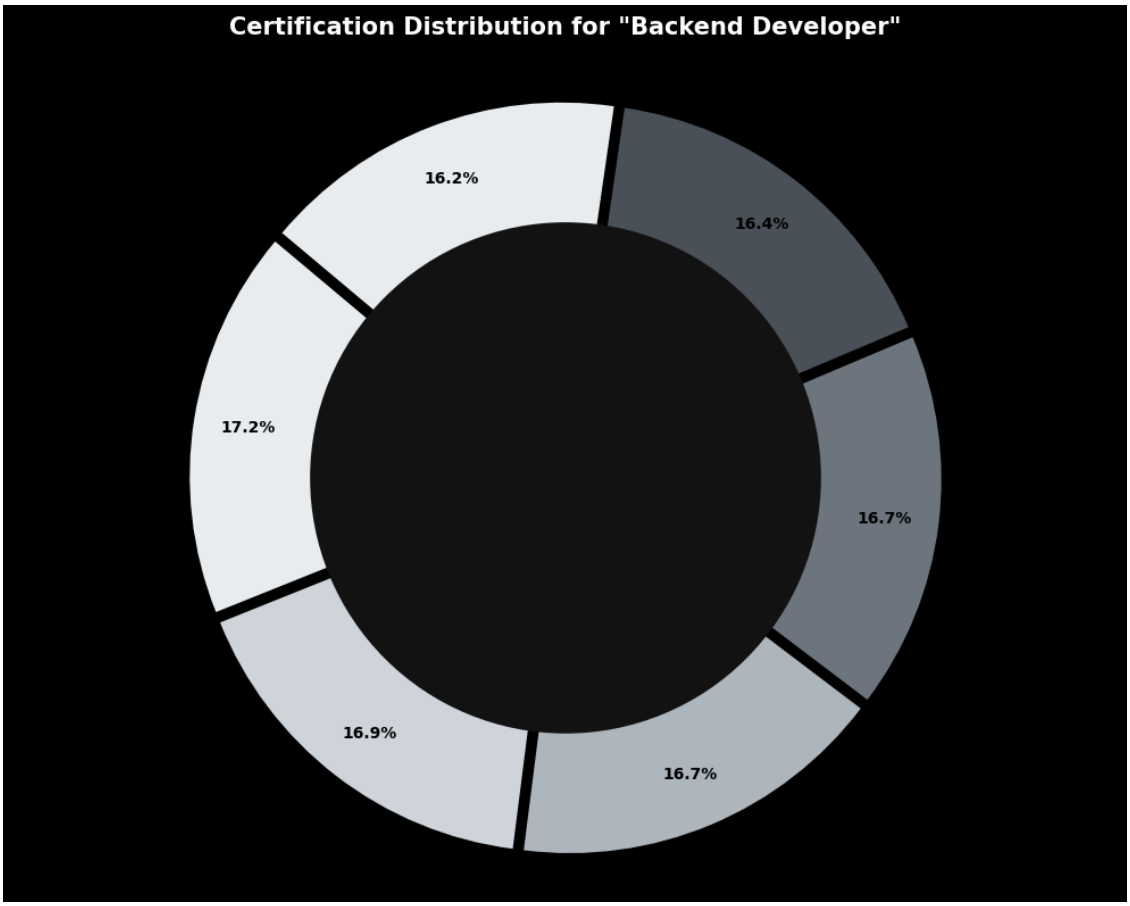
[47]:
```
cert_pivot = pd.crosstab(df['job_title'], df['certifications'])

print("Sample Counts of Certifications per Job Title:")
display(cert_pivot)
```

```
Sample Counts of Certifications per Job Title:
```

certifications	0	1	2	3	4	5
job_title						
AI Engineer	3481	3442	3491	3624	3501	3406
Backend Developer	3630	3418	3527	3573	3454	3523
Business Analyst	3447	3411	3467	3427	3468	3428
Cloud Engineer	3455	3483	3319	3465	3486	3478
Cybersecurity Analyst	3571	3493	3516	3544	3441	3394
Data Analyst	3455	3400	3426	3531	3439	3471
Data Scientist	3493	3402	3524	3415	3461	3595
DevOps Engineer	3536	3563	3501	3512	3421	3356
Frontend Developer	3446	3430	3422	3484	3424	3427
Machine Learning Engineer	3455	3428	3625	3448	3348	3373
Product Manager	3583	3538	3442	3493	3466	3428

```
Software Engineer          3463  3566  3484  3458  3558  3347
```

[48]:
```
backend_data = df[df['job_title'] == 'Backend Developer']
cert_counts = backend_data['certifications'].value_counts().reset_index()
cert_counts.columns = ['Certification', 'Count']

plt.style.use('dark_background')

steel_colors = ['#e9ecef', '#ced4da', '#adb5bd', '#6c757d', '#495057']

plt.pie(cert_counts['Count'],
        labels=cert_counts['Certification'],
        autopct='%1.1f%%',
        startangle=140,
        colors=steel_colors[:len(cert_counts)],
        pctdistance=0.85,
        explode=[0.03]*len(cert_counts),
        textprops={'color':"black", 'weight':"bold"})

centre_circle = plt.Circle((0,0), 0.70, fc='#121212')
fig = plt.gcf()
fig.set_size_inches(10, 8)
fig.gca().add_artist(centre_circle)

plt.title('Certification Distribution for "Backend Developer"', fontsize=15,␣
 ↪fontweight='bold', pad=20)
plt.axis('equal')
plt.tight_layout()
plt.show()
```



A striking and unusual symmetry was observed in the sample distribution across the dataset. Not only was the number of samples identical for each set of certificates, but this same pattern was repeated across all job titles.

Conclusion:

This strongly supports the hypothesis that this data may have been artificially generated.